One Sentence at a Time: A Quantitative History of Rationality in Economic Thought

ARTICLE HISTORY
Compiled December 18, 2025

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Abstract
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KEYWORDS
rationality; economics; bibliometrics; text mining; natural language processing; large language models

JEL CLASSIFICATION
B00; B20

Interactive application to explore the data

We have developed an interactive application to explore the data and results presented in this paper. It is available at: https://019adac8-81d4-aa0e-808c-08861c261fd2.share.connect.posit.cloud/. The first tab allows to explore the bibliometric networks; the second tab is a global summary of textual clusters.

Introduction

In the last decade, the number of history of economics papers employing quantitative methods has increased (see e.g., Goutsmedt, Claveau, and Herfeld 2023 special issue). Several essays have adopted a reflexive stance, examining the implications of using quantitative methods in the history of economics (Cherrier and Svorenčík 2018; Edwards, Giraud, and Schinckus 2018). These contributions have aimed to provide broad discussions on the use of such methods. However, given the virtually unlimited variety of quantitative approaches potentially relevant to historians of economics—and the wide array of research questions they can address—these reflections often remain abstract and offer little practical guidance. As a result, they tend to provide useful broad overviews, but without clearly articulating what quantification entails in practice, and how it can be both meaningful and challenging to integrate into research specific to the history and philosophy of economics.

Our contribution sits between a historical study that uses quantitative methods to answer a specific question and a general methodological discussion of those methods. From the collection of data to the interpretation of results, we illustrate concretely how these methods are useful and examine, in practice, the methodological choices they entail. We present a step-by-step application of specific quantitative methods to a broad topic: the history of rationality in the twentieth century.

We focus our discussion on unsupervised methods that have been particularly influential in the history of economics. Unsupervised methods are machine-learning techniques in which algorithms identify patterns from unlabelled data, as opposed to supervised learning methods—such as regressions—that learn patterns from labelled data to make inferences. The goal of unsupervised methods is not necessarily to provide a "measure" (Grimmer, Roberts, and Stewart 2022)—though they can be adapted to do so—but to organise in categories large corpora and enable accelerated and "distant reading" (Moretti 2013; Guldi 2023). They are well suited to historical inquiry because their unsupervised nature reduces the risk of presentism: rather than imposing present-day categories on the past, unsupervised algorithms uncover patterns directly from the data. When time is explicitly incorporated, they allow researchers to map the discipline at different points of time, to identify the emergence and decline of subjects and concepts, and to assess the influence of specific economists or ideas.

Our article uses two types of data, texts and citations. The analysis of textual corpora offers a direct window into the semantic content of debates, while citation data provides a lens through which to view intellectual interconnections and channels of influence within the discipline. We show how a large corpus of documents can be classified based on the information contained in textual and citation data—what we call "semantic clusters" and "bibliometric communities."[1]

Above all, our discussion aims to illustrate a core principle for historical inquiry with such unsupervised quantitative methods. They require continuous back-and-forth between aggregate quantitative results, complementary indicators used for interpretation, and preliminary knowledge and close reading of primary and secondary sources. These methods function not only as a form of corroboration but also as "discovery methods" (Grimmer, Roberts, and Stewart 2022): they facilitate the exploration of large datasets to reveal historical patterns. They often confirm, and sometimes complement, established findings. They can also reveal pitfalls and blind spots in existing research.

The concept of "rationality" is an effective focus for a concrete demonstration. First, many historians of economics engage with it in one way or another, which makes the exercise relevant for a broad audience. Second, because the concept is broad and pervasive in economics, applying quantitative methods to a very large corpus is particularly informative.[2] We use a corpus of around 290 000 articles in economics extending back to 1900 to show how these methods can handle long time horizons. Third, the multiple meanings attached to "rationality" and its uses applied to various subjects show how textual methods, coupled with bibliometrics, can help capture this semantic plurality.

In what follows, we concentrate on what such a quantitative analysis requires in practice. We highlight what kinds of questions and challenges arise at each stage of the research process. This focus lets us highlight (a) the crucial issue of selecting sources and building data; (b) how even simple quantitative assessments can be informative; (c) the challenges and subjective choices involved in building and adapting tools; and (d) why interpreting results demands careful attention to various indicators and close knowledge of both the corpus and

[1] The bibliometric communities, identified through network analysis, could also be called clusters. But we opted for "communities" in order to distinguish them from the "semantic clusters".

[2] To be clear, we don't think that quantitative methods are only helpful for very large corpora. However, it is where their surplus value may appear as the most obvious.

its historical context. By tracing, step by step, how we assemble and analyze our corpus, we aim to provide a practical example and a reflection on broader methodological issues faced by historians of economics in quantitative inquiries.

From sources to corpus

From sources to data

Discussions on quantitative methods often focus on the varieties of existing methods. In practice, however, analysis and interpretation come last. Most effort goes into collecting, cleaning, and structuring data, much of which remains invisible in published work. Sources rarely arrive as ready-to-use datasets; they must be transformed from somewhat raw materials into usable corpora. In short, the quantitative historian must be as much a data wrangler as a data analyst.

Bibliometric databases are a convenient way to access corpora of economic texts: they record relatively well-structured data, gathering key features of scientific output—such as authors, journals and affiliations, *etc.* Some of these databases, such as *Web of Science, OpenAlex,* or *Scopus,* record citation data, which enables tracing intellectual influence through diffusion patterns, and mapping scholarly contributions over time.

Each database has its own strengths and weaknesses. While at a very general level Web of Science, OpenAlex, or Scopus have similar coverage (Martín-Martín et al. 2021; Culbert et al. 2025), some studies might suffer from choosing a less appropriate database regarding the research questions. For instance, Scopus has a lower coverage of the top 5 economics journals before the 1990s and while being openaccess, OpenAlex has been less curated than the products of private publishers. Whatever the provider, less successful or now-defunct journals are also more likely to have incomplete or discontinuous digital coverage, impeding their inclusion for historical analyses. Additionally, databases are oriented toward English-speaking contributions; these databases may thus be deficient for a research project targeting another or multiple languages.[3] More generally, citation practices have only standardised progressively in the postwar period. Consequently, citation data are most of the time relatively poor before the 1960s. Last but not least, although they provide useful metadata and citation information, these databases generally lack full text, which is subject to copyright and therefore cannot be obtained from a single publisher.

Even when citations and full texts are available, the amount and quality of information that can be reliably encoded remain limited. Citation data are notoriously difficult to structure, because both the very notion of what constitutes a reference and the conventions governing its recording have changed over time—for instance, from references embedded in footnotes to the development of standardized bibliographies and the author–date system (Grafton 1999). Full texts face similar limitations: mathematical expressions and empirical material, such as tables, are often poorly captured or inconsistently encoded by providers. These shortcomings restrict what can be studied by quantitative tools.

A first important point is, therefore, that data availability shapes what can be asked and so answered. The scarcity and structure of available data affect every stage of inquiry, from the choice of research questions to the interpretations of quantitative results. In practice, research may be shaped as much by the scarcity of our data as by researchers' own preferences. It is of course possible to build handmade datasets from scratch, but, in the case of textual and citation data, such tasks remain time-consuming beyond small-scale study.

[3] One of the main challenges for a quantitative history of economics is to move beyond reliance on proprietary digital libraries and to actively develop open, historically inclusive corpora. This includes incorporating materials produced in non-Anglophone countries and expanding the range of textual formats considered—such as books, book reviews, working papers, and conference proceedings.

More commonly, it is often necessary to combine information from different databases to fulfill a specific goal. For example, a medium-scale study of the publications of the European Economic Review (Goutsmedt and Truc 2023)—few thousands documents—required to combine three heterogenous databases: Econlit (for JEL codes classification) Web of Science, and Scopus (due to imperfect coverage of Web of Science).

For our study of rationality, the first step was to identify the best sources. For citation data, the Web of Science (WoS) provides the most reliable and consistent coverage for our period and has been widely used in the history of economics. As for full text, we relied on three providers. First, JSTOR's full-text collection offers high-quality scans of most leading economics journals. Second, we used Scopus to identify peer-reviewed economics journals not included in JSTOR and to compile a complementary list of articles; for these, we retrieved full texts first through the Elsevier Full-Text API, when available. Third, remaining full texts were obtained through the ISTEX project, which provides access to a substantial corpus of documents for researchers affiliated with French universities.[4]

The Figure 1 shows the distribution of this corpus. It can be already noted that our corpus disproportionately represents Anglo-Saxon journals, which are also those most systematically digitized and preserved. Not only do we tend to overlook other traditions of research, but this also raises a serious issue of presentism: the fact that retrospective data on these journals are easily accessible today does not imply that they were equally central in earlier ones, nor that articles were the dominant vehicle for the diffusion of ideas. Rather, it reflects the fact that they had the resources and opportunities to digitize and maintain comprehensive digital archives of their publications.

Once access to full text is secured, another crucial step is to transform them into a usable database. While WoS citations are already delivered in a relatively structured form, full texts require substantial processing. In most cases, data are not given but result from a process that involves cleaning, categorizing, and selectively removing or reformatting information from raw sources in order to produce a dataset suitable for computational analysis. This challenge is particularly true in the history of economics, where the primary sources are the texts themselves—often unstructured and interspersed with various layers of textual content, such as section headings, footnotes, bibliographic references, or editorial annotations. This requires making important choices well before the actual stage of analysis.

Our first choice was to restrict the analysis to English-language articles, since cross-language comparisons involve additional challenges, which would go far beyond the scope of this article.[5] We also restrain our analysis to research articles and filter out book reviews, comments, editorial reports or obituaries.[6] While such materials can illuminate how rationality was debated, they are not primary sites for articulating new ideas within the discipline. Moreover, textual data are by nature voluminous, and filtering improves tractability for large-scale computation. At the document level, we focused on the body text and removed (as far as possible) peripheral elements such as acknowledgements, references, or appendices. We also focus on natural-language text and remove other information that is poorly OCR-processed and often unusable, such as mathematical formulas and data tables.

This data wrangling is not a tedious prelude to "real" historical analysis. Manipulating, cleaning, and structuring raw sources is a fundamental and often generative stage of research. As Lemercier and Zalc (2019, 62) reminds us, collecting and categorizing sources is also "a

[4]For more detailed information on the collection of data, see the appendix.

[5]Indeed, most textual methods are designed to identify commonalities within texts that share a specific linguistic structure. Hence, the same topic discussed by two documents in two different languages will likely not be identified as related because linguistic differences mask underlying semantic similarity.

[6]This filtering draws on JSTOR's and Scopus' own classifications, supplemented by some additional filtering from ourselves. Despite these safeguards, a perfectly clean restriction to research articles was not feasible, and some residual non-article items likely remain.

moment to reflect on the sources and the purpose of the research." Direct engagement with raw data can prompt the reevaluation of initial hypotheses and the emergence of new questions. In our case, preparing full-text documents raises choices about what counts as a relevant economic text. Should we include book reviews, working papers, or conference proceedings? How should we define an "economics journal"—restrict it to core outlets or extend it to interdisciplinary venues where economics appears regularly? Even within a single article, boundaries are ambiguous: should abstracts, footnotes, or appendices be analyzed or excluded? Each decision may carry historiographical implications. It shapes the corpus and, ultimately, the history of rationality our methods can reveal. There are rarely definitive and uncontestable choices, but rather a series of trade-offs that should be made explicit and required to engage with some of the material available.

To give an illustration, it is only after a first batch of exploratory analysis that we notice that some important economics journals were missing in JSTOR. If it was obvious from preliminary results that the emergence of behavioral economics impacted how economists discussed rationality, from our own expertise on the matter, we observed that some journals like the *Journal of Economic Behavior & Organization* or the *Journal of Behavioral Economics*—later to be the *Journal of Socio-Economics*—were missing in places where they should be predominant. This prompted us to further explore potential biases in JSTOR and to augment our corpus with new full texts extracted from Elsevier and the ISTEX project. Exploring raw sources is thus an important step that requires to already engage with your material and the existing literature.

The use of several databases also raised specific issues. Here, our goal was to combine our textual data (from the three providers mentioned above) with citation data from WoS. Achieving this required linking documents across datasets, yet WoS—like many databases—does not provide DOIs, which would otherwise serve as convenient unique identifiers. It therefore fell to us to develop matching procedures to determine whether an article retrieved from JSTOR or Scopus corresponded to the same article indexed in WoS. However, matching based solely on the title is unreliable. For example, the title *"Inflation and Unemployment"* may refer either to James Tobin's (1972) AEA presidential lecture or to Milton Friedman's (1977) Nobel lecture. Consequently, we had to supplement title information with additional metadata, including the journal name—which required standardizing journal titles across databases—as well as the publication year, volume, issue, and page range. Differences in journal coverage and in data storage practices (particularly title formatting) mean that it is generally impossible to achieve a complete match between databases. In our case, approximately one-quarter of the full-text documents after 1945 could not be matched to WoS records, which prevents us from analyzing their citation data.[7]

From data to corpus

In parallel with the transformation of sources into data, we also needed to establish the boundaries of our corpus, a process that involves selecting materials either *ex ante*, when choosing which sources to include, or *ex post*, when filtering the collected data. In the context of our project on the history of rationality in economics, this required us first to determine what qualifies as "economics," and second to identify which documents or parts of documents can be considered "texts" about rationality.

The first challenge was thus to define the extent of our corpus a priori, i.e. to delineate economics as an object of study. Some research objects are relatively easier to delineate, and the transition from a database to a well-defined corpus is therefore straightforward. For example, writing the history of a particular journal (Edwards 2020; Charles et al. 2025) or of one or several individuals (Truc 2025; Delcey 2025) entails comparatively definitional or

[7] See the appendix for more information on the matching process.

boundary challenges. While some large-scale studies focus on the discipline as a whole (Claveau and Gingras 2016; Truc et al. 2023; Ambrosino et al. 2018), such work still requires an operational definition of what are documents in "economics." This definitional issue becomes even more pronounced when the object of study is a specific "field" (Cherrier, this issue) .

Non-quantitative historical methods tend to address large objects by focusing on their "core" rather than their boundaries: historians of economics typically reconstruct the history of materials that are unambiguously part of the object under study, thus making a narrow definition often unnecessary. Quantitative approaches, however, require a well-defined corpus and consequently oblige researchers to impose simple, clear-cut boundaries on complex and ambivalent categories such as "economics". Defining the relevant historical materials therefore requires the adoption of specific conventions to approximate the object under study (Desrosières 2011). Defining a corpus constitutes a quantification convention—one that must always be assessed instrumentally, not as an absolute definition but as an operational hypothesis serving a specific research question.

Many proxies have been used in the history and philosophy of economics to define disciplines and sub-disciplines. For instance, (Goutsmedt and Truc 2023) used the JEL codes to select macroeconomic documents in a corpus of well-established economics journals, while (Truc 2022) and (Jullien and Truc 2024) relied respectively on citations data and institutional affiliations to identify the boundaries of behavioral economics. To choose a particular proxy, it is important to understand how it relates to the object under study. Both JEL codes and journal classifications—like JSTOR or Scopus classifications—can structure a corpus in ways that reflect their own histories, and researchers must therefore consider how these proxies shape the boundaries of their material. For example, while the JEL codes for neuroeconomics emerged around the same time as the first publications in the field, the JEL codes for behavioral economics appeared more than two decades after the earliest contributions (Truc 2023). A thorough knowledge of the history of the object studied is thus crucial for evaluating the adequacy and representativeness of the resulting corpus.

In our case, we first define economic documents by building the corpus from peer-review journals. This convention has both advantages and limitations. On the one hand, this approach is unambiguous, as it relies on agreeing upon a predefined list of journals, rather than, for instance, determining at the document level which individual articles qualify as economics. Moreover, journals constitute one of the main legitimate institutional markers of a discipline, alongside training programs and professional associations. On the other hand, focusing on journals means excluding other publication outlets, such as books or working papers.[8] In addition, it struggles to capture interdisciplinarity, whether in the form of interdisciplinary journals or of economists publishing in other disciplinary journals. Finally, this strategy requires the use of arbitrary criteria to decide which journals to include and which to exclude—particularly for journals at the margins of "economics", either because of their interdisciplinary orientation or because of uncertainty regarding their peer-review standards or academic practices. We eventually settled on a carefully curated list of 329 journals identified as economics journals in JSTOR and Scopus, but excluding journals that are not entirely academic, insufficiently focused on economics, or only founded within the last twenty years.[9] We were able to retrieve 289538 articles published in these journals from 1900 to 2009.

The second challenge was to restrain our corpus to documents engaging with the issue of

[8]Adding these other outlets would pose additional challenges. For both working papers and, above all, books, no large-scale databases provide comprehensive full texts or complete reference lists. In addition, working papers raise the problem of potential double counting, as they may eventually be published in economic journals.

[9]We did not filter journals by language a priori. Indeed, many non-anglo-saxon journals also publish English articles at some more or less frequent occasions. It was thus easier to filter by language a posteriori and at the document level, once articles were collected.

rationality. One of the most straightforward proxies are keywords (Truc 2023). Based on a predefined list of target terms (often called a "dictionary"), this approach restricts a corpus to documents that mention these terms with at least a given frequency. In addition to its relative simplicity, it works well for specific and unambiguous terms, like "stagflation" (Goutsmedt 2021) or "Agent-Based Models" (Baccini et al. 2025). But this approach may display several limitations when it is not the case. Indeed, it focuses on spotting occurrences of a term rather than a general idea. Just think about the various ways rationality could be discussed in the history of economics: beyond the term "rationality", economists employ various expressions that refer to close ideas such as maximising profits or expected utility, the *homo oeconomicus*, or the transitivity and completeness of preferences.

Going beyond simply searching for occurrences of "rationality" and "rational," we could have constructed an extensive dictionary of terms associated with the concept of rationality. This task requires substantial knowledge of the history of economics and of debates surrounding rationality, and thus presupposes that researchers possess adequate historical and conceptual background. Despite this, it remains difficult to prevent the dictionary-building process from introducing biases—for instance, by omitting concepts that are historically relevant but salient only during specific periods (such as "hedonism"), or by creating a disproportionate dictionary, with many terms related, for instance, to decision theory but few pertaining to macroeconomics or public economics. Consequently, this approach also constrains the potential for discovery: by setting the boundaries of the dictionary in advance, researchers may unintentionally exclude terms or themes of which they were unaware, and thus remain unaware of them throughout the analysis.

To overcome this issue, we use a Large Language Model (LLM) to identify documents—and in particular specific sentences within documents—that deal with rationality in our corpus of economic articles.[10] LLMs are trained on extremely large collections of text to learn patterns in language. In simple terms, they learn to predict missing words in a sentence or to determine whether two sentences follow each other. Through this training, the model develops billions of internal parameters that capture regularities in vocabulary, grammar, and meaning. When we input text into such a model, it translates sentences into vectors that summarize their context and meaning. This enables us to compare sentences not by the exact words they use but by their underlying ideas. For example, the same term—such as "model"—may carry different meanings depending on the surrounding context (*fashion model* vs *scientific model)* and a LLM can detect these variations. By converting words and sentences into vectors that reflect their semantic usage, LLMs make it possible to treat ideas and conceptual shifts as measurable objects, thereby opening new possibilities for the quantitative study of economic thought.

We rely on Sentence-BERT (Reimers and Gurevych 2019), an LLM designed specifically to produce sentence embeddings, that is, numerical vectors that represent the meaning of a sentence. The model assigns one vector to each sentence, which allows for straightforward semantic comparison: sentences that express similar ideas end up with vectors that are mathematically close to each other. Using Sentence-BERT, we vectorized more than 61 million sentences published between 1900 and 2009 from our database. Starting from sentences including "rationality" and "rational", we then searched for the sentences most similar to these anchor sentences. Thus, if sentence A explicitly uses the word "rationality," and its vector is close to that of sentence B—which never mentions the term—sentence B likely discusses a related idea.

Our approach does not simply consist of finding sentences that are close to the words "rationality" and "rational" over a 110-year period, though. Because our project is historical,

[10]Our goal here is not to provide the reader with extensive details on what these models are and how we use them (see the appendix for additional details and references), but rather to provide general intuitions about the use of the LLMs, to understand the building of our corpus.

it is crucial to consider how LLMs themselves handle historical language. LLMs are trained on billions of texts, but the overwhelming majority of these texts are recent.[11] As a result, they tend to offer a "presentist," numerical view of language. For example, current models struggle to reproduce writing styles from earlier periods and cannot reliably infer the publication date of a text (Underwood, Nelson, and Wilkens 2025). Sentence-BERT is subject to the same limitations, and the sentence embeddings it produces inevitably reflect this bias. Besides, as illustrated by Figure 1, our corpus is exponential according to years and most sentences come from recent articles. Taken together, these effects make it very likely that the sentences the model identifies as closest to "rationality" will predominantly come from recent articles. To limit this presentist effect, we adapted the way we compare sentences over time. Instead of comparing a sentence from, say, 1910 directly to a "representative vector" of all sentences in our corpus that contain the words "rationality" or "rational," we constructed a series of representative vectors. For each year, we build a moving average vector, with a five-year symmetric window, of all the sentences mentioning "rationality" or "rational". Concretely, for the year 1910, we averaged the vectors of all sentences containing "rationality" or "rational" from 1905 to 1915. This produces a period-specific reference point that reflects how these terms were used at that particular moment in time. A sentence from 1910 is therefore judged similar not to the general, and likely modern-day meaning of rationality, but to the way the concept was expressed during its own historical period. This helps ensure that our analysis is sensitive to historical changes in language. Since each document is a set of sentences, we also identify the documents—and the authors—that discuss rationality most intensively or frequently. Table 1 and Table 2 illustrate respectively the closest sentence and documents from our representative vectors in 1910.

Exploring the corpus

Simple exploration

Before returning to our specific corpus on rationality and diving into more advanced LLM-based analyses, we first step back and examine simpler quantitative methods applied to our broader set of economics articles. Indeed, quantitative approaches come in many forms and levels of complexity. In fact, quantification does not need to be sophisticated to be useful. Simple indicators may not always provide the clearest answers to research questions, but they can play an important exploratory role: helping researchers refine their questions, identify anomalies, or detect unexpected patterns.

For textual data, one of the most straightforward indicators is term frequency, that is counting how often particular words or expressions appear. This metric has been used repeatedly in the literature, especially to track the emergence or decline of fields within economics. In well-delimited domains, term frequency can serve as a reliable proxy for intellectual dynamics. For instance, Truc (2023) shows that counting occurrences of highly specific neuroeconomics terms—such as "striatum" or "prefrontal"—closely approximates more advanced quantitative measures, and thus provides a simple but meaningful signal of activity in the field.

Figure 2 shows the relative frequency (with respect to the total number of words published each year) of "rational" and "rationality" in our corpus. The figure reveals a steady postwar increase of the use of both terms, likely reflecting the progressive consolidation of rational choice theory in economics. We observe a marked surge after the 1970s, with a pronounced peak for "rational" in the 1980s, and a smaller peak for "rationality" in the 1990s. The rise of

[11] Moreover, the texts used to train these models are not primarily academic articles in economics. This means that there may be important gaps between the general language patterns the model has learned and the specific vocabulary, concepts, and writing practices of our "domain" (see e.g. Zhang, Rojcek, and Leippold 2025).

"rational expectations" in macroeconomics and the emergence of behavioral economics following the publication of Kahneman and Tversky (1979) likely contributed to this pattern. Of course, we cannot easily infer the precise drivers of these trends from such simple metrics. However, the clear upward movement signals an important moment in the evolution of rationality within economics—at least in terms of textual intensity—thereby motivating more targeted forms of investigation.

While our usage of a large language model will shed light on the context surrounding this rise, such context can also be approximated with simpler, more straightforward measures. Co-occurrence analysis helps recover the different intellectual settings in which rationality is invoked. Figure 3 reports, by decade, the five words most frequently adjacent to "rational" or "rationality," showing how these associations shift over time. Before the 1930s, the picture was heterogeneous. The notion was tied to the marginalist idea of "calculation," but it referred not only to individual behavior but also to "systems" or "organizations." The discussion was explicitly methodological, as indicated by the prominence of terms such as "method," "explanation," "law," and "foundation." From the 1940s onward, the rise of choice theory places rationality at the center of economic modeling as a device for describing and formalizing behavior. This is mostly visible with the rise of multiple common bi-grams (i.e., combination of two words) that remain stable from the 1930s through the 1980s such as "economic rationality", "rational choice", "rational behavior". By the 1970s—and especially the 1980s—the framework was both extended and contested. First, the most common bi-gram by far becomes "rational expectations" signalling the emergence of a new predominant concept extending rationality. In the 1980s, "rational expectations" appeared 15 more times than the other most common bi-grams. Second, while the concept "bounded rationality" was developed by Herbert Simon initially in the 1950s, the concept only became prevalent during the 1990s with the bi-gram becoming the fourth most common one.

Beyond textual data, citation data also constitute a useful source of information for corpus exploration. The evolution of an idea depends not only on how it is formulated by its authors, but also on how it is appropriated, and reinterpreted by readers. A substantial literature in citation theory examines how citations function as a scientific practice and how they should be interpreted. As emphasized in this literature (see, e.g., **TahamtanCore1979?**), citations may reflect a wide range of motivations—from a genuine desire to acknowledge intellectual debt to more strategic uses aimed at persuading readers or satisfying referees. Nevertheless, citations at the very least signal a relationship that can help trace the lineage and diffusion of ideas, even when the reasons behind the citation are not purely intellectual.

Counting the citations of specific documents offers a first, simple approach to assessing the impact of an author or a particular publication. There are clear reasons to incorporate citation studies into historical research. Beyond tracing diffusion, highly cited papers tend to be more visible and attract greater engagement—a dynamic that reflects the well-known Matthew effect (**mertonMatthew1968?**). Recent work also shows that citations influence reading behavior and perceptions of quality: highly cited papers are more likely to be read thoroughly and treated as significant intellectual contributions (Teplitskiy et al. 2022).

Historians routinely discuss scientific influence and recognition using proxies such as major grants, prizes, and honors. For example, Sent's (2004) narrative of the transition from the dominance of rational choice, through the limited success of "old" behavioral economics (e.g., Simon, George Katona), to the emergence of "new" behavioral economics is organized around such milestones. In this sense, citations serve as a complementary proxy alongside these traditional indicators. For instance, although both Simon and Kahneman received the Nobel Prize, Kahneman ultimately exerted a broader influence on the discipline—something that is reflected in citation patterns. In a large qualitative–quantitative study of the Nobel Prize, Offer and Soderberg (2016) distinguished several trajectories among laureates: those who peak at the prize and subsequently decline, "innovators with staying power," "still rising" winners honored before their citation peak, and late winners recognized long after their peak.

Figure 4 plots citation patterns across all WoS economics journals and across the top five journals for four seminal references that critiqued the standard approach to rationality in economics. Simon (1955) and Allais (1953) represent early critiques of neoclassical rational choice, while Akerlof (1970) and Kahneman and Tversky (1979) are foundational contributions to what economists now refer to as "new" behavioral economics.

The influence of "new" behavioral economics clearly exceeds that of the earlier tradition, both in the top five journals and in the broader set of economics journals indexed in Web of Science. Whereas Simon (1955) and Allais (1953) are cited by no more than 0.2% of all economics articles, Kahneman and Tversky (1979) surpasses 1% by the late 2010s and continues to rise. The contrast lies not only in the scale of influence but also in the speed and timing of acceptance: citations to the two "new" behavioral papers grow rapidly and steadily from the moment of publication, while Simon and Allais reach their modest peaks only around the time of their Nobel Prizes—or even later in the 2000s, with the renewed attention generated by "new" behavioral economics. As argued by Offer and Soderberg (2016), many laureates experience a "Nobel premium," a modest increase in citations following the award. Simon and Allais conform to this pattern, but Kahneman and Tversky represent an exceptional case: after the Nobel, their previously declining citation trajectory reverses and climbs sharply throughout the period studied (Offer and Soderberg 2016).

More advanced exploration

Citation counts are a blunt tool and cannot answer many questions of interest to historians of economics: Who cites these works? Are they cited together? In what intellectual contexts do they appear? Moreover, focusing on a small set of references introduces a degree of arbitrariness into the analysis. Likewise, simply counting the words that appear next to "rationality" tells us little about whether these words are used jointly within the same argument or whether they belong to distinct topics and contexts that mobilize the concept differently. It also overlooks the much broader vocabulary related to rationality (e.g., profit maximization, expected utility, social choice).

Most recent quantitative studies in the history of economics rely on what can be grouped under the label of "unsupervised methods." These approaches apply algorithms to a corpus—whether full texts or citation data—to generate classifications and assign categories that are not predetermined by the researcher but that emerge from statistical patterns in the data itself, which gives the approach its "unsupervised" character.[12] Assigning such categories imposes a form of internal organization on large and otherwise unwieldy bodies of material, enabling an accelerated or "distant reading" (Moretti 2013). The typical output of these methods is a map of a discipline or research area, showing how different entities—topics, articles, authors—relate to one another.

The counterpart of unsupervised methods are supervised methods—such as regressions—which estimate predefined relationships in the data. In contrast, unsupervised methods are particularly well suited to exploratory analysis. In this research context, analysts often lack precise knowledge of the corpus and its various intellectual contexts. While supervised methods require clearly specified hypotheses and well-defined variables, unsupervised approaches allow researchers to let a *chosen* structure of the material emerge. This is especially true for large-scale and long-run analysis, where labelled data become harder to obtain and increase the risk of imposing categories that are inadequate or anachronistic relative to the historical context. That does not mean, however, that an

[12]"Unsupervised" does not mean that the modeller has no influence on the output. The analysts must choose a particular model and its parameters and these choices embed methodological priors about the types of patterns the procedure is likely to reveal. For instance, one famous crucial parameter in many unsupervised models is the number of categories that will emerge from the algorithm.

unsupervised algorithm can be applied blindly to historical questions: a substantial part of our effort has consisted in adapting the method to our historical data, especially to accommodate the growth of the corpus over time.

In this paper, for citation data, we use a method commonly employed in the history of economic thought: bibliographic coupling [ref]. We begin with a corpus composed of the 10% of articles whose average embeddings are most similar to our annual representative vectors for "rationality" and "rational." In this network, articles are represented as nodes, and connections between them depend on the number of shared references in their bibliographies. The more references two articles share, the stronger their link—and the closer they appear in a two-dimensional network visualization. The underlying premise is that shared references serve as a proxy for intellectual proximity. These maps help delineate the boundaries of disciplines, fields, or sub-specialties and reveal their internal organization, including hierarchies and core–periphery structures.

However, citation practices tend to favor more recent works, which makes it unhelpful for historians to construct a single citation network spanning an entire century. Following a method that has proven effective in the field [three papers], we therefore split our corpus into overlapping eight-year windows, beginning in 1960 (1960–1967; 1961–1968; … ; 2002–2009), and built a series of 43 separate networks. Using community-detection algorithms (**traag?**…), we identify groups of documents that share a substantial fraction of references and therefore have a similar intellectual background; we refer to these groups as "bibliographic communities." Our next step is to identify communities that persist over time. When two communities from consecutive networks share many of the same articles, we treat them as instances of the same underlying group—an "intertemporal bibliographic community." In this way, our bibliographic communities bring together documents from different periods of time but that are likely to engage with rationality in similar ways, based on their shared citation patterns. This method allows us both to *zoom in* on specific communities at a given period and to *zoom out* by reconstructing the broader picture of a dynamic intellectual field.

For textual data, historians of economics frequently rely on topic modeling [Ref]. Topic modeling identifies the latent themes in a corpus by learning distributions of words associated with each topic. Each document is then represented by a mixture of these topics—often with one or two dominating—while each topic can be described through its most representative words and expressions. While topic modeling is a convenient tool for historical analysis, our research question pushed us toward a more targeted approach.

Because we focus on one large and historically fluid concept, we draw on the literature on semantic drift [ref] and take inspiration from (**ref?**). Rather than starting with full documents, we filter each year's corpus to retain only the top 1% of sentences that are most similar to our annual representative vectors of "rationality" and "rational." In other words, we extract the sentences most likely to engage with the concept of rationality in any form.[13] As with bibliographic coupling, we want a method that groups together sentences that employ similar meanings of rationality. We therefore cluster the sentence embeddings using the HDBSCAN algorithm, a widely used unsupervised method for clustering LLM-generated vectors.

Again, historical perspective matters. Because the distribution of identified sentences is strongly present-biased (Figure 1), clustering all sentences at once would risk over-representing recent periods. We therefore perform clustering separately for each decade from 1900 onward (merging the first two decades due to fewer sentences).[14] As in our

[13] Taking the top 1% means that we consider, once excluded as much as we can, bibliographies, headings, etc, that 1 sentence over 100 in economics articles deal with rationality. This threshold is, of course, open to discussion and can easily be adjusted.

[14] Note here that we don't use overlapping windows in this case, notably for computational reasons: finding HDSBCAN clusters on a set of tens of thousands of sentences is much computationally intensive than finding

bibliographic approach, we then seek to form larger groups over time, allowing us to "zoom in" and "zoom out" depending on what we are searching for. Using the cosine similarity between clusters across decades, we merge the closest ones into 17 "intertemporal semantic clusters."[15]

With both methods, we thus obtain a list of "intertemporal bibliographic communities" and "intertemporal semantic clusters." These allow us to identify subgroups within our corpus across multiple dimensions: different subfields of economics (e.g., behavioral economics, macroeconomics), different conceptions of rationality (e.g., bounded rationality, rational expectations), or even, more heuristically, different methodological traditions (e.g., experimental work, econometrics). Both approaches have clear strengths and limitations. Some are largely inherited from their respective sources. In our case, citation data cannot be meaningfully exploited before the 1960s, while our OCR full texts handle formulas and tables poorly, making economic models—an essential component of our story—less visible in the semantic analysis than through citations. Choosing between the two therefore depends directly on the specific historical questions one aims to address.

Ideally, relying on *both* methods serves complementary purposes. First, comparing the results of different unsupervised computational approaches—such as LLM-based embeddings and bibliometric analysis—acts as a form of robustness check.[16] The systematic comparison of sources and methods is a long-standing principle in historical research, allowing scholars to triangulate information rather than depend on a single perspective. Second, the two methods illuminate different aspects of intellectual history. Full texts reveal the meanings, contexts, and semantic nuances of rationality as economists used the concept in their writing, independently of their institutional proximity or citation habits. Citation data, by contrast, highlight patterns of intellectual influence and diffusion: who cites whom, how ideas travel across fields, and where intellectual boundaries lie. It conveys sometimes a more sociological dimension: scholars tend to cite the work of people publishing in similar journals, going to similar conferences, etc. The following sub-section illustrates how we interpret our results, and how the two methods complementarity may be useful.

Interpreting "Results"

Taken together, the textual and bibliometric approaches allow us to study both the semantic and thematic contexts of the uses of rationality in economics and the channels through which these ideas circulated within the discipline. But how should such a "study" proceed? Unlike simpler quantitative tools, unsupervised methods do not produce ready-made results. In our case, they generate statistical categories—bibliographic communities and semantic clusters—that lack predefined conceptual "labels". This strategy—reducing a large and complex corpus to a smaller number of coherent groups—is well established. Yet the groups themselves are created solely on the basis of statistical similarities, and their historical or conceptual significance emerges only through subsequent analysis. Groups resulting from an unsupervised method can comprise, depending on the corpus, anything from a few dozen to several thousand texts. To make sense of what brings peculiar texts together, we therefore need indicators that identify shared features and help hierarchise the documents to be read first. At the same time, a good knowledge of the corpus and of the relevant intellectual debates is indispensable for making sense of the raw computational results, even once they have been informed by a set of indicators.

Thus, whether at the stage of data construction or in the interpretation of statistical

bibliographic communities for at most five thousands of articles..

[15] Refer to the appendix for more details about this.

[16] Modeling choices (for example, the minimum size of clusters in the HDBSCAN algorithm) affect the partition in communities and clusters. Since these outputs are never self-interpreting and always require human qualitative assessment, it is time consuming to conduct robustness checks.

patterns, quantitative methods demand a continuous dialogue with qualitative interpretation, supported by a close reading of the secondary literature on the topic at stake. Only through this back-and-forth can we turn algorithmic groupings into meaningful historical insights. This is both a weakness and a strength. On the one hand, the analysis is not immediately transparent, as it requires the construction of intermediate tools—such as our interactive applications—to guide interpretation. On the other hand, it is consistent with the practices of historians of economic thought, who likewise select, prioritise, and read texts in order to make sense of their corpus.

Let us take as an example the semantic cluster 11, which gathers close to 20,000 sentences from 1920 to 2009. How can we get a sense of what this cluster actually coalesces around? Before any qualitative interpretation or contextualization with the secondary literature is possible, we must first produce a series of indicators to characterize the cluster. Such indicators include:

- The most identifying words of the cluster for each decade. While the first period of the cluster (the 1920s) deals notably with "rationalisation" and "human reason," the 1950s are more directly focused on "rationalism," "rationality," and "rational behavior." After the 1970s and until 2009, "bounded rationality" emerges as a core concept in this cluster.
- The most recurring authors. While Talcott Parsons and Frank Knight appear frequently in the 1920s and 1930s, Herbert Simon becomes a central author in the 1950s and 1970s. In the 1990s, we notably find Robert Sugden.
- The journals that contain the most sentences associated with this cluster. The *Journal of Economic Issues*, oriented toward evolutionary economics, figures prominently, as do more behavioral-economics journals such as the *Journal of Socio-Economics* and the *Journal of Economic Behavior and Organization.*
- The sentences within the semantic cluster that are closest to its year-representative vector, i.e., those closest to "rational" and "rationality," as well as the most representative sentences of the cluster for each period. The first category tends to highlight how the concept of rationality is discussed, while the second category centers more directly on the core semantic content of the cluster, often offering clues about the topics and debates that animate it in different periods. For instance, in the 1950s, one of the sentences closest to the representative vectors is Simon's claim that "If man, according to this interpretation, makes decisions and choices that have some appearance of rationality, rationality in real life must involve something simpler than maximization of utility or profit" (Simon 1959, 259–60). As for the most representative sentences, Allan Drazen (Drazen 1980, 294) warned in the 1980s that "Deciding on the suitable notion or definition of rationality is not an easy task," in his survey of the disequilibrium-theory literature in macroeconomics.
- The articles with the most sentences in the cluster for each decade. Here we find, for example, Terence Hutchison's "Expectation and Rational Conduct" (Hutchison 1937); Jacob Marschak's "Rational Behavior, Uncertain Prospects, and Measurable Utility" (Marschak 1950); Herbert Simon's 1978 Richard T. Ely Lecture at the AEA meetings (Simon 1978) and his Nobel lecture (Simon 1979); as well as Sugden's 1991 *Economic Journal* survey on rational choice (Sugden 1991).
- For the periods after the 1950s, the most cited references (based on the number of citations per article, whether the article has one or ten sentences in the cluster). In the 1950s, for example, one of the most important references within this cluster is John von Neumann and Oskar Morgenstern's *Theory of Games and Economic Behavior* (Von Neumann and Morgenstern 1944), while Kahneman and Tversky (1979) becomes the most cited reference in the 2000s.

With all these indicators—and prior to any further qualitative investigation—we can already see that this semantic cluster is centrally relevant to our study, as it bears directly on the concept of rationality itself. Our textual approach is thus able to extract some specific uses of rationality, here debates on the meaning of rationality itself in economics, and to see the evolution of these uses across time but also across topics and subdisciplines (in game theory, macroeconomics, behavioral economics or evolutionary economics).

The bibliometric approach does not allow for such fine-grained temporal or cross-subdisciplinary exploration. In our bibliometric communities, no grouping addresses rationality and its limits in general across multiple decades. Rather, bibliometric analysis identifies groups of texts from authors in close academic communities. In relation to semantic cluster 11, we observe, for instance, the emergence of a "Behavioral Economics and Choice Theory" community in the 1977–1984 window. As with the semantic clusters, for each bibliometric community in each temporal window we extract, notably, the sentences closest to the representative vectors—based on the articles belonging to the community—as well as the most cited references and the most identifying words. We also examine how each community in a given window originates (i.e., whether it derives primarily from the same or from different communities in preceding periods) and what it becomes in the following period (i.e., its subsequent "destiny").

Discussion

Equipped with these groupings and series of indicators, we can begin to make sense of our results. How can they help enrich, complete, and refine our understanding of the various and evolving meanings of rationality in economics? This paper does not propose an alternative history of the concept, which would extend far beyond the scope of this methodological discussion. Rather, we highlight selected findings that corroborate and strengthen strands of the existing literature, notably by allowing us to make claims about the prevalence and timing of considerations about rationality, while also broadening the scope of that literature and opening avenues for further research.

Throughout much of the period—but especially before the 1940s—sentences captured by our representative vectors refer sometimes less to the rationality of economic agents than to the rational character of economists' arguments, understood in terms of coherence or sound reasoning. This pattern reflects the logic of our semantic approach: LLMs tend to associate explicit mentions of rationality with broader discussions of reasoning and coherence, even when the connection to later, behavior-centered notions of rationality remains indirect. We retain this ambivalence deliberately since it reveals both historical transformation and the interpretative challenges inherent to large-scale analysis.

Our goal is also to show how we navigate the results and triangulate the indicators in order to produce coherent historical narratives. While we rely on the secondary literature, the articles we foreground are only those identified by our indicators as some of the most significant for understanding the semantic clusters or bibliometric communities.[17] We provide two complementary discussions: a macro-level perspective that highlights broad patterns in the evolution of rationality, and a micro-level analysis that focuses on the emergence of a specific topic—behavioral economics.

[17] Of course, as in any historical inquiry of this kind, we are not entirely protected from selective emphasis and from choosing illustrations that support a particular line of interpretation—all the more so given that the scope of our study is too broad to summarize every pattern revealed by our results. Other scholars could have, at various junctures, pursued different interpretive directions on the basis of the same material. The publication of the interactive applications we used to explore the results thus serves as an exercise in transparency and—in addition to being, we hope, useful for historians of economic thought more generally—allows readers to assess the robustness and limits of our narrative.

Through the Telescope: Broad Patterns on the History of Rationality

When our quantitative inquiry begins in 1900, the concept of rationality had already gained prominence, in line with the rise of scientific and abstract reasoning in the late nineteenth century. The scope of rational behavior was closely tied to debates on the proper domain of economic analysis (Giocoli 2003).

With the marginalists, the economic agent was increasingly defined by calculative capacity. Jevons, for instance, portrayed the agent as a "calculating machine" balancing pleasures and pains, a view later formalized as utility maximization (Maas 1999).

In our first two decades, cluster 14 represents discussion centred on utility and brings together insights on marginal utility from, among others, Sidney. J. Chapman, Arthur Pigou, Francis Y. Edgeworth, and John M. Clark. Cluster 4 also captures discussions about utility: economic theory is debated through the history of economic thought. Arguments about utility were framed as interpretations, critiques, or extensions of classical authors—most notably Mill, Ricardo or Malthus. Yet, if anything, marginal utility theory appears mostly under severe criticism in this period, especially from institutionalist economists such as Thorstein Veblen, Wesley Mitchell, John Maurice Clark, and Rexford Tugwell.[18]

The criticism was directed along two main lines. First, as Giocoli (2003, 48–50) points out, institutionalist economists rejected the "unfounded psychological underpinnings of neoclassical economics" and sought to replace them with sounder foundations (see also Yonay 2001, 101–6). They proposed instead to renew the economic method by drawing on psychology and its emerging "behaviorist" approach (Rutherford 2001, 175–76). Semantic cluster 9—concerned throughout the period with general methodological issues in economics—captures a significant share of sentences from 1900–1920 that articulate such institutionalist critiques. For instance, Veblen (1909, 622–24) attacked the "marginal-utility school" for relying on the simplistic depiction of human behavior implied by the "hedonistic calculus." Institutionalists were energized by new developments in psychology. In his review of the literature on "Human Behavior and Economics," Mitchell emphasized the growing interest of economists in psychological matters, which stemmed from "a somewhat tardy recognition that hedonism is unsound psychology, and that the economics of both Ricardo and Jevons originally rested on hedonistic preconceptions" (Mitchell 1914, 1). For economics, the crucial issue was thus to choose "between providing a sounder psychological basis for our analysis, and holding that its psychological basis does not concern the economist" (2).[19] Such criticism did not falter in the 1920s. In cluster 11—centered at the time on rationalisation and human reason—Tugwell (1922, 317) encouraged economists to move beyond the "rigid classical *homo economicus*" and noted that "psychologists have already pretty well revolutionized the scientific definition of human nature."

Second, for institutionalist economists, institutions were indispensable for understanding economic behavior, and different institutions implied different behaviors; there was no single, context-independent *homo oeconomicus*. In cluster 9, Veblen (1909, 620) again argued against the "statical character" of marginal-utility theory, which offered "no theory of a movement of any kind," being occupied solely with "the adjustment of values to a given situation." Similarly, E. H. Downey (1910, 253), in the cluster 14, claimed that "Marginal-utility economics has nothing to say of the genesis, growth, or current working of economic institutions" (see also Rutherford 2013, 141). Its "deliverances," he argued, were therefore "futile for the problem of social betterment," since such "betterment" concerned

[18]However, because we focus on English, published articles, our clusters are likely biased toward American journals, notably in the first decades.

[19]Some years earlier, Mitchell (1910)[97; see semantic cluster 9 and 4] already pushed forward similar claims, regretting that "few economists have regarded the study of psychology as a necessary part of the equipment of their work", often relying on "tacit preconceptions" rather than to seek to "psychologists to gain a knowledge of the mind and its modes of operation".

the adjustment of institutions to changing circumstances and ideals. In the early twentieth century, economic behavior, according to institutionalists, had to be understood through the prism of the "pecuniary institutions" that shaped it. For instance, Charles Cooley (Cooley 1915) underlined that "pecuniary valuation," or economic valuation, is only possible because of the institutions that make such valuation possible (see also Rutherford 2013, 58–59). Figure Figure 3 shows well the importance of "pecuniary" as a neighbor to "rational" and "rationality" in the first twenty years of the twentieth century.

The centrality of institutionalist economists—as well as the prominence of U.S. journals in our corpus—is visible in the scope of the semantic clusters in the first periods: sentences related to rationality, in addition to the clusters already mentioned, were grouped into themes related to the railway industry (semantic cluster 8), progressive taxation (cluster 17), tariffs and commercial policy in the United States (cluster 8 again and part of cluster 2), or agriculture (cluster 13). In line with institutionalism, most of these economists examined these themes through the laws and rules governing economic activity, focusing on how institutions structure the behaviour and interests of collective actors rather than on abstract individual assumptions. In these clusters, "rationality" is rarely mentioned explicitly, even if we find discussions on the behavior of "business men" (cluster 12), "workmen" (cluster 8) or "owners of land" and "farmers" (cluster 13). These clusters gradually decline in importance, and most of them disappear in the postwar period. Indeed, the period from 1930 to 1959 displays substantial transformations in the structure of debates concerning rationality.

The most obvious transformations are those described by Giocoli (2003) as "the escape from psychology" and, more generally, the shift from a "system-of-forces"—"economic processes generated by market and non-market forces"—to a "system-of-relations," in which the aim becomes to establish the existence and properties of equilibria through the "validation and mutual consistency of given formal conditions" (Giocoli 2003, 4–5). "Rationality" was progressively distinguished from "reason" or "intelligence" and recast as a formal, axiomatic notion—"rigid rules that determine unique solutions" (Erickson et al. 2013; see also Klaes and Sent 2005).

While cluster 14 (on utility) constituted only a tiny share of sentences between 1900 and 1919, and was absent in the 1920s, it gains importance in the 1930s to the 1950s (though still representing less than 5% of all sentences). During this period, the heart of the debate concerned the measurability of utility and the opposition between "cardinal" and "ordinal" utility. In the 1930s, the debate still bore on the psychological dimension of utility measurement. In 1933, John Hicks and Roy Allen (1934) coauthored an article on demand analysis that eliminated marginal utility by appealing to the concept of the marginal rate of substitution—the slope of the indifference curve (see Moscati 2019, 98–100). Already in 1932, Allen had made this project explicit: by starting from "preferential discrimination," or individuals' preferences over different bundles of goods, it becomes unnecessary to "make any assumption about the existence of 'total utility' or about the measurability of 'utility'; the hedonistic hypothesis has been rendered superfluous" (Allen 1932, 207). In other words, "Subjective and psychological concepts have been discarded from pure economic theory" (ibid.).

Yet this position was not uncontested. Indeed, in these debates, terms such as "introspection," "satisfaction(s)," and "pleasure" recur frequently in the cluster during the 1930s and 1940s (see for instance Armstrong 1939). But they disappear from the top identifying words in the 1950s. During this period, debates on ordinal versus cardinal utility continued—with a renewed defense of the cardinalist position and ongoing disputes about the very meaning of "measuring utility" (Moscati 2019, chap. 10)—yet these discussions had largely emancipated themselves from psychological or behaviorist considerations. These debates are also associated with a new cluster (cluster 5), which appears in the 1940s and grows substantially (to nearly 8% of all sentences) in the 1950s. This cluster concerns the formalization of agents' preferences and choices. Here, we encounter what would become the standard vocabulary of decision theory: “individual orderings,” “preferences,” “decisions,” “transitive,” etc., and Von Neumann and Morgenstern (1944), Friedman and Savage (1948), and Savage (1954) appear as the most cited references in the core articles of the cluster. To complete this panorama, a game-theory cluster emerges in the 1950s, which ultimately comes to represent nearly 15% of all sentences after 1990.

Transformations in the conception of rationality are also visible through clusters that remain thematically stable over time, yet reflect profound shifts in the discipline’s methodology and the growing centrality of rational choice theory. Semantic cluster 12, for instance, which spans the period from 1900 to 1979, deals with firms and entrepreneurs. In the 1920s, the cluster emphasizes entrepreneurial decision-making, understood primarily as prudence facing uncertainty rather than formal optimization. From the 1940s, this cluster increasingly centers on “profit maximization.” The cluster captures the core of the so-called “marginalist controversy” (Backhouse 2009), notably through Richard Lester’s attack on marginal theory and Fritz Machlup’s (1946) response. Machlup mounted a defense of profit maximization and, more broadly, of marginalist theory. He rejected the evidential value of Lester’s questionnaire-based surveys of entrepreneurs, while Lester replied that “at the heart of economic theory should be an adequate analysis and understanding of the psychology, policies, and practices of business management in modern industry” (Lester 1947, 146).[20] The marginalist controversy more generally provided a key intellectual background for Friedman’s essay on positive economics (1953), in which he formulated the famous “as if” principle to justify, among other assumptions, profit maximization. Interestingly, in the 1950s the most cited articles within cluster 12 were Machlup’s (1946) and Friedman’s (Friedman and Savage 1948), both of which relied heavily on “as if” reasoning, notably to justify the assumption of expected utility.

Shifts in the discipline’s methodology are also captured by semantic cluster 9, which gathers sentences in which authors reflect on their own methodological choices. While in the 1920s and 1930s economists still engaged with notions such as “human nature,” “passions,” or “prejudices,” the postwar period saw the growing prominence of terms such as “logical implications” and “assumptions,” followed, from the 1970s onward, by an increasing focus on “models” and “optimality.”

It thus becomes clear that by the 1950s the conception of rationality had been profoundly transformed, moving in a more formal direction and largely emancipated from psychology. Another transformation also deserves emphasis: as rationality was recast in terms of consistent choice rather than as a psychological portrait of *homo oeconomicus*, the normative dimension expanded—from individual behavior to questions of social choice and to the design of “rationalizing” policies (Herfeld 2018; Hands 2015).

The emergence of this social dimension of rationality is already observable in the interwar period. In the 1930s, cluster 11 focused on the concept—relatively neglected in the history of economic thought—of “rationalization.” In a literature review on this concept, Robert Brady (Brady 1932, 526) remarked that “scarcely any term… has occasioned more discussion and dispute.” The concept referred to “every technique, program, or plan of organization which promised to promote (…) the ‘efficiency’ of individual enterprises, entire industries, or even the total of economic processes nationally or internationally considered.” Attention to “rationalisation” is also visible in cluster 13, devoted to agricultural economics and the rise of “farm management” aimed at improving farmers’ living conditions. It becomes even more

[20] Machlup’s position was that even if a “goodly portion of all business behavior may be non-rational, thoughtless, [or] blindly repetitive” (1946, 520), questionnaire evidence and empirical findings more generally did not demonstrate the failure of marginal theory when properly interpreted. In a similar defense of profit maximization, Leonid Hurwicz (1946, 110) first acknowledged that it was not “inconceivable that business is run more by routine than by rationality”, but argued that behavior may appear irrational or merely “routine” from the standpoint of a purely static theory that ignores uncertainty, while proving fully “rational” once uncertainty and long-run effects are taken into account (see also Herfeld 2018).

pronounced with the multiplication of debates on economic planning and the rationalization of policy decisions from the 1940s onward, particularly in cluster 8.

As rationality increasingly came to be framed through modern microeconomics, the concept was extended to both individual and collective choice. In semantic cluster 5—gathering sentences on decision theory—the rational "decision making" is not applied not only for individuals but also for firms and public policy. It is also in this cluster that issues of social choice emerge, notably debates on the implications of Arrow's impossibility theorem (1951; see e.g. Igersheim 2019).[21] Arrow's contribution also stimulated the formation of cluster 16 on law and economics in the 1960s, where it appears as the primary reference, alongside works by Downs (1957) and Buchanan and Tullock (1962). This cluster is closely associated with publications by *Public Choice* (see Cherrier and Fleury 2017). Welfare economics and public choice, as well as public finance (Desmarais-Tremblay, Johnson, and Sturn 2023), are also clearly visible in the bibliometric analysis after 1960. Indeed, one of the major bibliometric communities of the 1960s is organized around public goods and externalities, with James Buchanan, Ronald Coase, and Richard Musgrave as central references. The prominence of public finance increases further in subsequent decades, with the appearance of another community devoted to optimal taxation after 1968; taken together, these two communities account for nearly one fifth of the entire network in this period.

We have seen how the meaning of rationality progressively transformed and formalized after the 1930s, and how the rationalization of collective decision-making became central in the postwar period. So far, however, we have said little about macroeconomics. Macroeconomics constituted a relatively important semantic cluster from the beginning of our period, in 1900, notably through monetary economics (cluster 1). This cluster grew in importance in the 1930s, particularly in connection with debates surrounding Keynes's work, liquidity preference, and unemployment. It is only in the 1970s, however, that this cluster becomes dominant in our panorama of rationality. This shift is closely linked to the emergence of the concept of "rational expectations," which profoundly transformed the way rationality was discussed in economics. The concept was developed by Muth (1961) but only later popularized when Carnegie colleagues—Robert E. Lucas, Edward C. Prescott, and Thomas J. Sargent—applied it to macroeconomics (Lucas 1972; Sargent and Wallace 1973). By the late 1970s, it had circulated beyond academia into policy circles and the press and was often described as a "rational expectations revolution" (Duarte 2025). Our methods help account for this rapid diffusion beyond controversial policy debates during the stagflation era. First, the results show no significant trace of rational expectations in the 1960s in either the bibliometric or textual outputs.[22] But from the 1970s onward, it occupies a central place in the literature on rationality, reaching its apogee in the 1980s, when it accounted for nearly 30% of all sentences in the textual analysis, as well as a comparable share of articles in the bibliometric analysis, spread across several bibliometric communities ("Rational Expectations and Business Cycles," "Exchange Rate Economics," and "Inflation, Expectations, and Interest Rates").

A similar trajectory to macroeconomics can also be observed in finance. The origins of financial issues can be traced to semantic cluster 3 which focused initially on the theory of investment. Until the 1930s, the issue was framed around the returns of capital. For instance, the cluster includes the so-called Hayek-Knight controversy on the conception of capital (**CohenHayek2003?**). After the 1940s, decision-making under uncertainty becomes a central theme, crystallized by works such as Shackle (1942) on investment under radical uncertainty, which figures among the most representative articles. One decade later, Jack

[21] This grouping of sentences in the 1970s actually took its origins in the semantic cluster 14 on utility, which ended in the 1960s, with a focus on welfare economics.

[22] This highlights a key caveat. Because these methods foreground statistically "significant" patterns and indicators' prevalence, they are not always well suited to tracing and interpreting the origins of a concept or theory with the care that close historical study and archival research provide.

(**hirshleiferInvestment1958?**) inscribed investment theory within the rising neoclassical conception of rationality, by formalizing investment decisions as a problem of intertemporal maximization of consumption. In the same vein, the Modigliani–Miller theorem (**ModiglianiCost1958?**) further consolidated this shift by grounding firm financial decisions as an arbitrage problem, opening the path to modern financial economics. Their theorem, they argued, can be used "as a basis for rational investment decision-making within the firm" (**ModiglianiCost1958?** 296). After the 1960s, cluster 3 became much more prominent, consistently accounting for more than 10% of all sentences. In this period, rationality is increasingly framed in formal decision-theoretic terms and focused on appropriate investment choices under uncertainty. Hence, our analysis shows that rationality in finance enters primarily through corporate finance, rather than through asset-pricing issues such as market efficiency, where rationality long remains an implicit assumption and rarely discussed explicitly prior to the emergence of behavioral finance.

From the 1980s onward, modern asset pricing becomes dominant within cluster 3 as indicated by the growing prominence of terms such as "assets," "arbitrage," and "capital asset pricing". We also observe increasing proximities between the finance literature and the rational-expectations framework from the mid-1970s onward. This is both visible in the semantic clusters and in the bibliometric communities, in particular, the "Rational Expectations and Market Information," which emerges in the 1976–1983 window. A central contribution found in both semantic and bibliometric clusters is Grossman and Stiglitz (1980), which draws on Lucas's imperfect-information framework (Lucas 1972) and combines rational expectations with noisy signals to question market efficiency. More generally, rational expectations models—initially developed in a macroeconomic context—became a central framework for analyzing the (ir)rational use of information in finance (see Delcey and Sergi 2023).

The 1970s and 1980s thus marked a relative dominance of macroeconomics and finance, with comparatively less weight given to debates on individual rationality. This situation begins to change gradually from the 1980s onward. In the 1979–1986 bibliometric network, a "Behavioral Decision Theory" community appears (representing around 2% of the network), capturing the early emergence of behavioral economics as promoted by Kahneman and Tversky. In the 1990s, the rise of behavioral economics—together with new approaches in game theory—fundamentally altered this configuration, leading to more specialized investigations of individual rationality, which now dominate and are distributed across multiple bibliometric communities. This transformation represents not merely a shift in research topics, but a profound reframing of how economics approaches rationality: from an assumption embedded in theories and models to an object of direct investigation at the individual level.

We have sought here to offer a broad panorama of the major transformations in the concept of "rationality" in economics. This approach allows us to identify large-scale changes when they become visible in the data—namely, when particular clusters come to represent a significant share of sentences (or of the bibliometric networks). However, historians of economic thought are often concerned with understanding how such transformations came about: which contributions, authors, and communities played a decisive role. Addressing these questions requires a more focused perspective, tracing the evolution of specific topics and literatures. This is the aim of the next section, which moves beyond the panoramic view to provide a more detailed narrative of the rise of behavioral economics outlined above.

Under the Microscope: Simon's Reception and the Birth of Behavioural Economics

The term "behavioral economics" has a complex history, with George Katona often credited as an early adopter (**giladEconomic1984?**; **hosseiniGeorge2011?**). Sent (2004) made an historical distinction between “old” and “new” behavioral economics to separates the “old” heterogenous attempts by Katona, Simon and others to reform mainstream economics from the “new” more homogenous heuristic and biases research program of Kahneman and Tversky. Sent argues that ”new” behavioral economics succeeded precisely because it worked within the existing paradigm rather than challenging it fundamentally. Additionally, it emerged in the 1980s when economic theory faced multiple epistemological challenges, creating an opportune moment for alternative approaches. While our method cannot pinpoint the exact epistemological reasons for this shift, we can trace the reception and influence of both approaches to understand how and when they diverged.

Simon's concept of ”bounded rationality” stands as one of the most influential critiques of standard rationality in economics in the sense that it framed a lot of the debates surrounding rationality. His presence dominated 1950s discussions, with his seminal articles and Nobel lecture (Simon 1979) appearing across numerous research areas from social choice theory to the theory of the firm and price setting.

Our textual analysis reveals Simon's conceptual centrality in some areas of economics. Textual cluster 11 about individual rationality in economics changed rapidly after the influence of Simon rised in the 1970s. While in the 1950s the discussion was structured by rationality terms (“rational”, “empirical reality” or “irrationality”), by the 1970s and after ”bounded rationality” became the cluster's most prevalent expression. By the 1990s, discussions of ”boundedly rational agents” and ”procedural rationality” clearly positioned Simon's concepts as central to economic thought signaling a late but significant integration. Not only did the way economists talked about rationality changed radically after the introduction of Simon’s ideas, but these discussoins became also more prevalent overall. While in the 1950s 5.76 % of all sentences belong to the topic this share increased to 7.85% in the 1980s. This is corroborated by citation analysis as citations to Simon (1955) only rised very slowly in economics after its publications with peak citations happening in the post 2000s (Figure 4).

However, Simon’s influence remain relatively limited overall. Textual cluster 11 peak in the 1990s with 9.56% of all sentence belonging to this topic, down to 7.43% in the early 2000s. Indeed, while Simon’s bounded rationality remain conceptually in parts of economics, there was a larger shift that happened and reshaped how economists approached rationality. Progressively after the 1980s, cluster 11 relative importance decreases in favor of more specialized clusters also about individual rationality. Cluster 5 and 15 about expected utility theory, risk and ambiguity. Cluster 6 and 7 about intertemporal behavior and other-regarding behavior. All these clusters become increasingly important in the 1980s with most being individually bigger than cluster 11 in share of sentences. While they do integrate bounded rationality ideas, they do so by using “new” behavioral economics concepts ranging from prospect theory for risk to the paradigmatic experimental results of the ultimatum game (**gualaParadigmatic2008?**).

More generally, Simon’s conceptual influence speardeaded by bounded rationality did not translate into stable research communities. Bibliometrically, only small, unstable clusters formed around Simon's work. In the 1960s and early 1970s, his ideas appeared in operations research circles (*Firm Behavior and Planning*) and organizational theory critiques of profit maximization (*Organizational Efficiency and Firm Theory*). Simon his more generally associated with other researcher critical perfect rationality like Winter (1971), which applies satisficing to firm behavior, and Cyert and George (1969). A related strand is Leibenstein (1966) on “X-efficiency,” which challenges economics’ focus on allocative efficiency. Together, Simon, Winter, Cyert (with George), and Leibenstein form a diverse critique of how rationality—especially through profit maximization—is employed in economics, from an organizational theory perspective. After 1969–1976, this stream landed in a smaller, short-lived community and references to Simon and critiques of profit maximization then remain scattered, moving through several small communities (e.g, cl_761, cl_251, *Welfare*

Economics and Public Finance, *Evolutionary Economic Theory*). While Simon's influence in economics was enduring, it remains marginal as a structuring force.

Like with textual clusters, the bibliometric analysis reveal a larger shift that absorb Simon’s legacy. In the 1979–1987 period, most clusters citing Simon became part of a large ”Behavioral Decision Theory” cluster alongside Kahneman and Tversky. Contrary to Simon’s, the Heuristics and Biases research program of Kahneman, Tversky and other “new” behavioral economists lead to multiple stable clusters that form well identified and large research communities. A main “Behavioral Decision Theory” stable cluster in the 1980s made of the early “new” behavioral ecomists (Kahneman, Tversky, Richard Thaler, George Loewenstein

The crystallization of a distinct ”Behavioral Decision Theory” around Kahneman and Tversky's contributions signal the beginning of an explosive growth of ”new” behavioral economics. The cluster spins off several autonomous communities capturing the emergence of sub-fields within behavioral economics: pro-social behavior (cl_x), intertemporal decision-making (cl_x), behavioral game theory (cl_x), and behavioral finance (cl_x). By the 2000s, behavioral economics had become the dominant venue for research on rationality, with five behavioral economics clusters representing approximately 40% of the entire network.

The contrast with Simon is striking. While ”bounded” and ”procedural” rationality remain regularly invoked concepts, no later research community clearly carries Simon's heritage as a bibliometric anchor. The scale and speed of acceptance differed dramatically between the two approaches. By 1980, Kahneman and Tversky (1979) was already more cited than Simon (1955) had been at that time (Figure 4). By 1985, prospect theory had surpassed the lifetime citation peak that Simon's work ever achieved in economics. (Figure 4). By the 2000s, most microeconomics publications about rationality are related to behavioral economics, and more generally, most publications about rationality are connected to the research program.

A first surprising result from our quantitative study is the stark contrasting reception of Kahneman and Simon in terms of temporality. For historian Floris Heukelom (**heukelomSense2012?**; Heukelom 2014), a pivotal moment in the history of behavioral economics is the explicit creation of research program within the Sloan and Russell Sage Foundations between 1984 and 1992 that most notably led to the pairing of Kahneman and Thaler that changed the trajectory of behavioral economics. However, our quantitative analysis suggests that even in the early 1980s it was clear that something particular was happening in terms of reception with the work of Kahneman and Tversky. Despite Kahneman and Tversky (1979) being the only economics publication of duo until 1985 and long before the Sloan-Sage Foundation research program (1984–1992) that formalized behavioral economics, prospect theory became very cited rapidly (Figure 4). As early as the 1977–1984 bibliometirc network, we find a cluster structured by the work of Kahneman and Tversky suggesting that not only their article his cited, but it rapidly structured a new strand of research in a way that never happened with Simon’s work in economics.

This is particularly remarkable given the external similarities between the two cases. Both Kahneman and Simon published in top economic journals and received Nobel Prizes. While Simon’s Nobel only increase his influence in economics marginally, Kahneman’s Nobel reversed a downward citation trend in the 2000s to an upward trend cimenting his contribution as a structural pillar of a larger wide spanning research program.

A second surprising result is that citations to Simon (1955) have never been higher than since the ascent of the new behavioral economics. Earl (2022) is critical of how much “old” behavioral economics seems to have been forgotten by “new” behavioral economists:

- “Neither Kahneman nor Thaler have sought to promote earlier behavioral economics alongside more recent work. Instead, they give the impression that behavioral economics started around 1979–1980 with the publication of Kahneman and Tversky’s

(1979) article on prospect theory and that theory's use by Thaler (1980). All in all, this is a very curious state of affairs: a cynic might suggest that it looks rather as if the earlier work has been airbrushed from the history of economic thought by the strategic redefinition of what constitutes behavioral economics. A more charitable and reflexive view would see the situation as resulting from insufficient familiarity with the earlier literature [...]" (Earl 2022, 2)

The rising citations to older work from Simon or Allais in our example show that while it is possible that most "new" behavioral economics rarely acknowledge the contributions of "old" behavioral economists, the scale of the rise of "new" behavioral economics still led to more attention paid to "old" behavioral economists than ever before in economics. This resurgence admits at least three interpretations. A more favorable reading is that the "new" program helped revive abandoned research directions and moved toward reconciliation with earlier strands (something promoted by Sent (2008) or more recently by Earl (2022)). A more unfavorable view sees intellectual appropriation, in which classic references are reframed to fit the new agenda, renewing interest but with a biased and presentist lens (Mongin 2019). A third more critical interpretation is that the citation increase reflects a growing backlash against the "new" program from the standpoint of "old" behavioral economics. Preliminary evidence supports the second interpretation. This is suggested by the sparse and unstructured citations patterns to "old" behavioral economics and the fact that discussions of rationality are increasingly framed by "new" behavioral and experimental concepts. While a definitive conclusion is beyond the scope of this paper, our study provides a roadmap for future quantitative and qualitative research needed to further the understanding of the relationship between "old" and "new" behavioral economics.

Our quantitative studies delineates the specific chronology and magnitude of the different attempts at stirring a behavioral shift in economics. However, as made clear by Figure 4, it is not just a linear story of a shift from one program to the other. citations to Simon (1955) have never been higher than since the ascent of the new behavioral economics. This resurgence admits at least three interpretations. A more favorable reading is that the "new" program helped revive abandoned research directions and moved toward reconciliation with earlier strands (something promoted by Sent (2008) or more recently by Earl (2022)). A more unfavorable view sees intellectual appropriation, in which classic references are reframed to fit the new agenda, renewing interest but with a biased and presentist lens (Mongin 2019). A third more critical interpretation is that the citation increase reflects a growing backlash against the "new" program from the standpoint of "old" behavioral economics. Preliminary evidence supports the second interpretation. This is suggested by the sparse and unstructured citations patterns to "old" behavioral economics and the fact that discussions of rationality are increasingly framed by "new" behavioral and experimental concepts. While a definitive conclusion is beyond the scope of this paper, our study provides a roadmap for future quantitative and qualitative research needed to further the understanding of the relationship between "old" and "new" behavioral economics.

Conclusion

This article is first and foremost methodological: it aims to demonstrate, through a concrete application, the usefulness of quantitative methods for the history of economic thought. Nonetheless, it does so by breaking new ground in the history and philosophy of economics. First, the corpus we build—comprising nearly 290,000 full-text economics articles spanning more than a century—is, to our knowledge, the largest and most comprehensive database ever used to study economic contributions in English. Second, this article constitutes the first application of quantitative methods to the study of semantic change within the history of economics, an area that represents an important and growing strand of the broader literature on quantitative text analysis (Periti and Tahmasebi 2024).

Our approach allows us to identify, within this corpus, the sentences most closely associated with the words “rational” and “rationality” and, by extension, the articles most likely to engage with rationality in one way or another. This procedure necessarily involves setting similarity thresholds to determine when a sentence or an article can be considered sufficiently “close.” By retaining approximately 500,000 sentences for the textual analysis and 29,000 articles for the bibliometric analysis, we deliberately opted for a broad initial corpus. Higher thresholds would have reduced these numbers and restricted the material to discussions more narrowly centered on rationality. Our methodological choice was instead to cast a wide net at the outset, in order to preserve flexibility for subsequent stages of analysis.

This choice is central to the interest of the unsupervised methods we deploy. By allowing the algorithms to identify dozens of semantic clusters and bibliometric communities within short temporal windows (eight years or a decade), and by then aggregating these groupings over time and characterizing them through multiple indicators, we can conduct analyses at different scales. This makes it possible both to *zoom out*, by examining the overall configuration of clusters across periods, and to *zoom in*, by focusing on subsets of clusters more directly concerned with rationality. As shown in the previous section, a panoramic view that considers all semantic clusters may initially include groups that appear only loosely connected to rationality—especially in the early decades. While examining these clusters can be informative for understanding the broader intellectual context, the analyst can also foreground more explicitly rationality-oriented clusters when addressing more targeted research questions. This flexibility is only possible because the initial corpus is sufficiently inclusive, reducing the risk of overlooking relevant developments. More generally, our approach allows the scale of analysis to be adjusted to the question at hand, rather than imposing it ex ante.

We want to conclude by drawing several general methodological and historiographic claims from the approaches developed in this article.

First, although quantitative methods in the history and philosophy of economics have been applied mainly to the postwar—and often post-1970s—period, our study shows that they can also be used effectively for earlier periods. Despite problems of text recognition in JSTOR articles prior to the 1930s, our methods nonetheless provide a fine-grained depiction of economic debates in the early twentieth century. In particular, we were able to recover most of the institutionalist contributions on rationality highlighted by Rutherford (2013) and Yonay (2001). At the same time, our results bring to light additional contributions not covered in these accounts, thereby opening avenues for further inquiry. Of course, some important works remain absent because they were published as books rather than journal articles. This limitation suggests that extending the corpus to include books would be a worthwhile direction for future research.

Second, a key property of groupings produced by unsupervised methods is that they bring together documents that share cognitive content without presupposing agreement. The texts clustered together address the same objects or problems, but they may do so in divergent, and sometimes opposing, ways. As a result, our methods make it possible to identify sites of resistance to dominant models or ideas. What emerges most clearly are the objects that attract sustained attention; within those sites, one can then examine who endorses, revises, or contests particular ways of treating them. This feature opens the way to a more “symmetrical” historical analysis (Bloor 1976), one that gives analytical weight not only to intellectual “winners” but also to those whose positions did not ultimately prevail.

Finally, these methods are also “discovery tools.” While they can be used to confirm or challenge existing claims in the history and philosophy of economics, they also help identify patterns that have so far been absent—or only marginally present—in the historical literature. We have briefly pointed to some of these patterns above. By making available two interactive applications to explore our results, we hope to encourage readers to pursue their

Table 1.: Sentences from the corpus closest to the representative vector in 1910.

Sentence	Cosine similarity
From another side, the concepts supply the basis for rationality.	0.690
The recital is important only as tending to show that a theory of valuation which places the emphasis upon rationalistic appraisal overlooks the most important features of the process which it seeks to explain.	0.686
Complete reliance may be placed upon the rationality of both the pecuniary and the hedonic subject.	0.678
It is exceedingly doubtful whether such hyper-rational view of human nature could have gained wide credence among men not themselves disciplined in the use of pecuniary concepts.	0.670
More fundamental is the great problem of accounting for economic rationality itself.	0.656

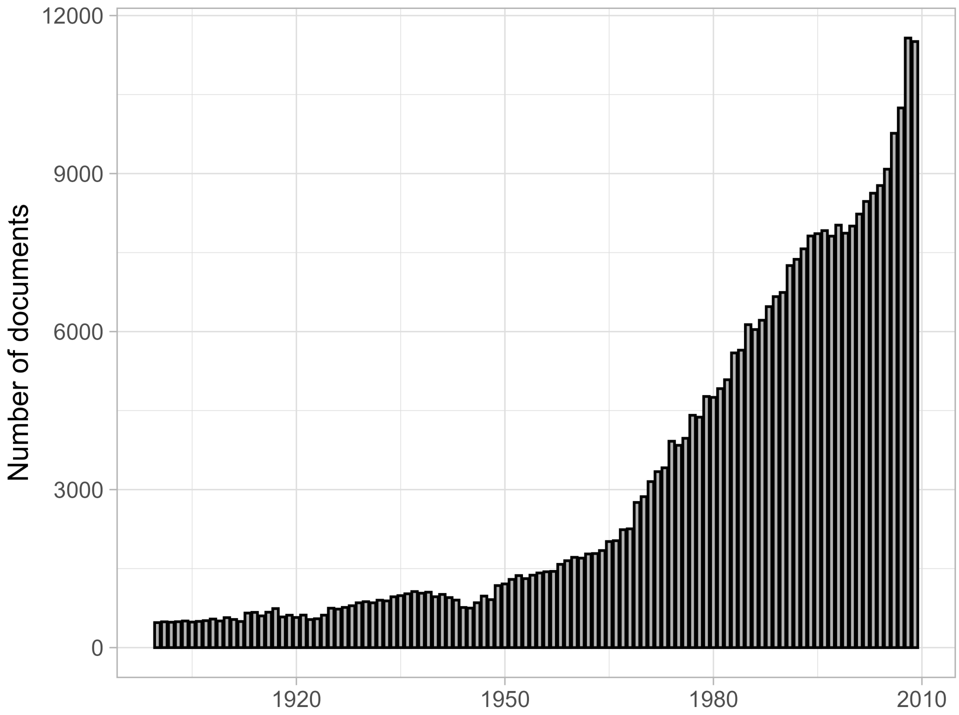
Table 2.: Documents from the corpus closest to the representative vector in 1910.

Title	Authors	Journal	Cosine similarity
The Rationality of Economic Activity	Wesley C. Mitchell	Journal of Political Economy	0.862
Social Productivity Versus Private Acquisition	H. J. Davenport	The Quarterly Journal of Economics	0.824
The Futility of Marginal Utility	E. H. Downey	Journal of Political Economy	0.786
Observation in Economics. Annual Address of the President	Davis R. Dewey	American Economic Association Quarterly	0.778
The Economics of Henry George's "Progress and Poverty"	Edgar H. Johnson	Journal of Political Economy	0.765

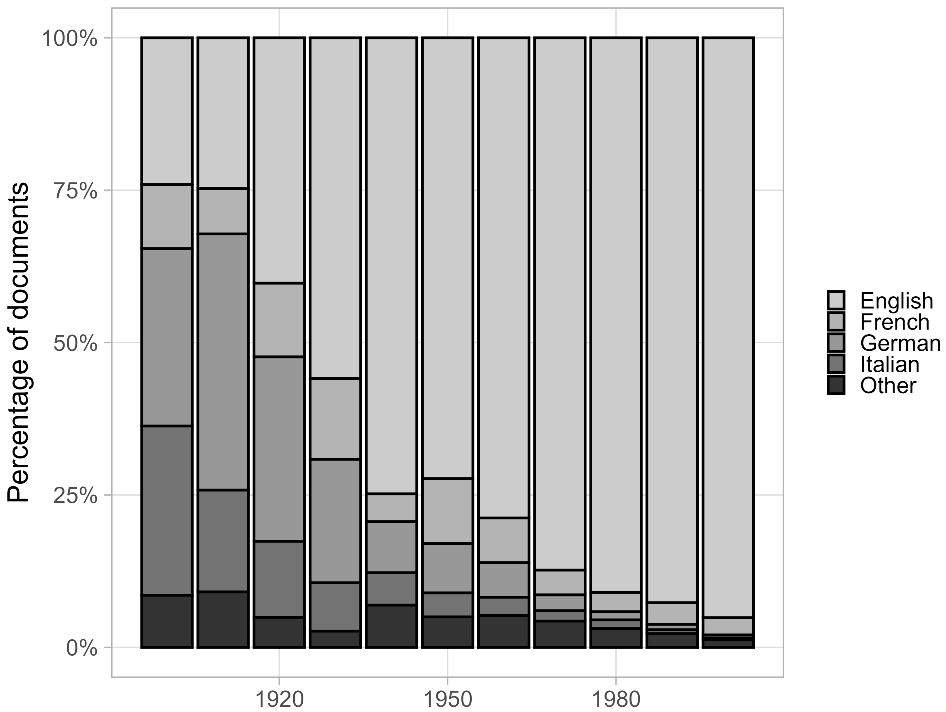
Note:
Documents vectors are computed as the average of their sentences vectors.

own paths of discovery within the corpus.

Figures



(a) Documents



(b) Languages

Figure 1.: Distribution of documents and languages in the full-text database over time (1900-2009).

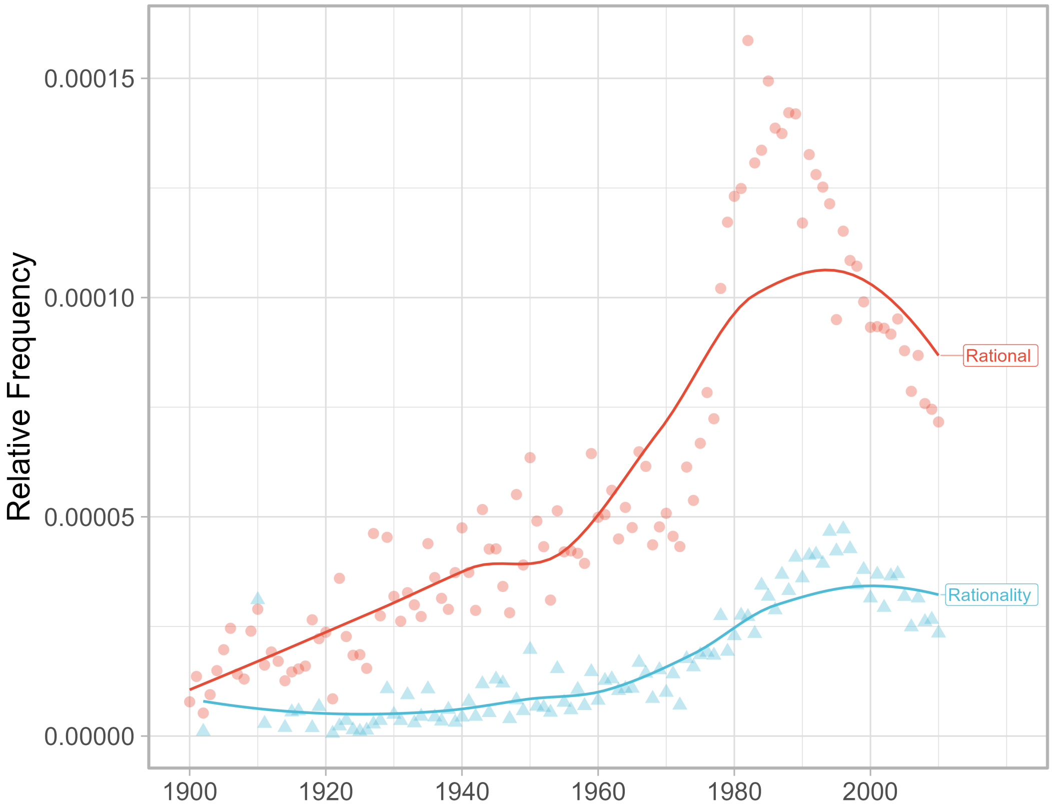


Figure 2.: Relative frequency of terms “rational” and “rationality” in the jstor database (1900-2009). Points are the empirical data. Lines are the loess smoothed trends.

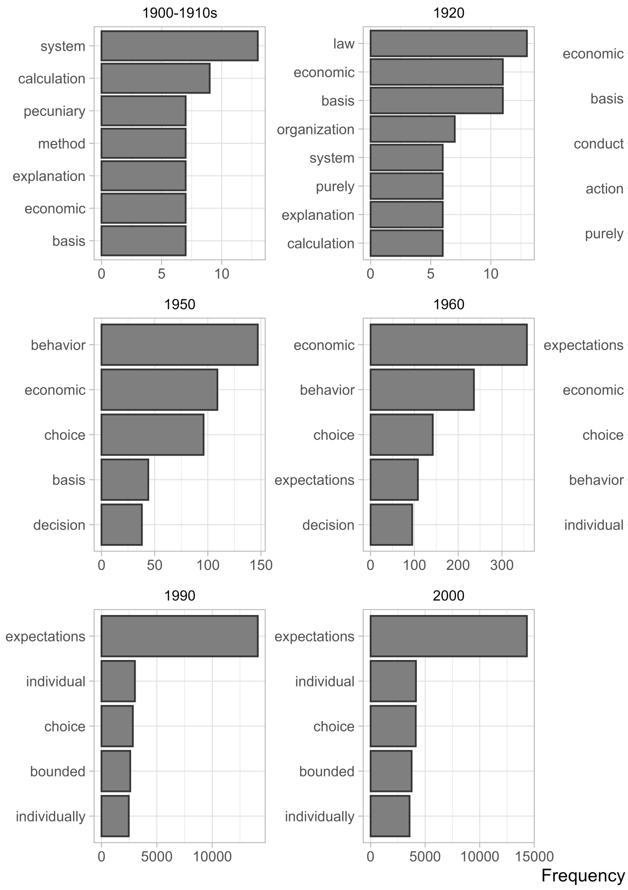


Figure 3.: Most frequent neighbor terms of “rational” and “rationality” by decade

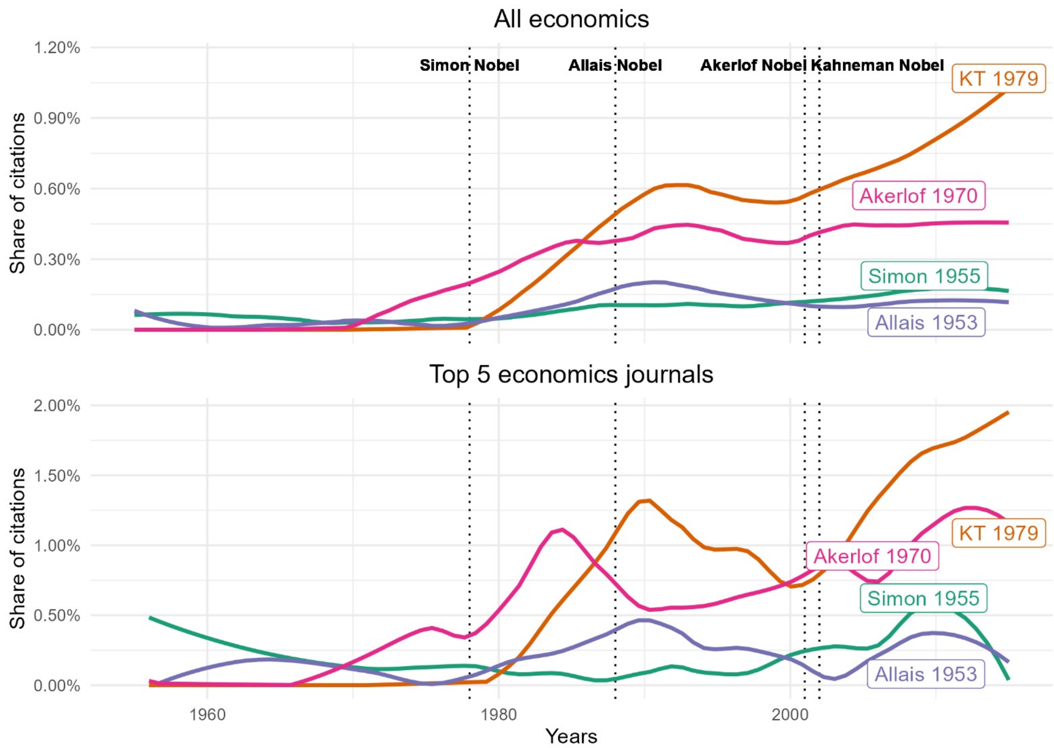


Figure 4.: Relative citation of a selection of seminal papers on behavioral economics

References

Akerlof, George A. 1970. "The Market for "Lemons": Quality Uncertainty and the Market Mechanism." *The Quarterly Journal of Economics* 84 (3): 488–500. https://doi.org/10.2307/1879431.

Allais, M. 1953. "Le Comportement de l'Homme Rationnel Devant Le Risque: Critique Des Postulats Et Axiomes de l'Ecole Americaine." *Econometrica* 21 (4): 503. https://doi.org/10.2307/1907921.

Allen, R. G. D. 1932. "The Foundations of a Mathematical Theory of Exchange." *Economica*, no. 36: 197–226. https://doi.org/10.2307/2549268.

Ambrosino, Angela, Mario Cedrini, John B. Davis, Stefano Fiori, Marco Guerzoni, and Massimiliano Nuccio. 2018. "What Topic Modeling Could Reveal about the Evolution of Economics." *Journal of Economic Methodology* 25 (4): 329–48. https://doi.org/10.1080/1350178X.2018.1529215.

Armstrong, W. E. 1939. "The Determinateness of the Utility Function." *The Economic Journal* 49 (195): 453–67. https://doi.org/10.2307/2224802.

Arrow, Kenneth J. 1951. "Social Choice and Individual Values." *Cowles Commission Papers* 12.

Baccini, Alberto, Martina Cioni, Muriel Dal Pont Legrand, and Eugenio Petrovich. 2025. "Does Cross-Fertilization Occur in Recent Macroeconomics? A Quantitative Exploration of the Interactions Between DSGE and MAB Models." *Journal of Economic Surveys* 39 (4): 1758–94. https://doi.org/10.1111/joes.12674.

Backhouse, Roger E. 2009. "Friedman's 1953 Essay and the Marginalist Controversy." In *The Methodology of Positive Economics: Reflections on the Milton Friedman Legacy*, 217–40. Cambridge University Press Cambridge.

Bloor, David. 1976. "The Strong Programme in the Sociology of Knowledge." *Knowledge and*

Social Imagery 2 (1976): 3–23.
Brady, Robert A. 1932. "The Meaning of Rationalization: An Analysis of the Literature." *The Quarterly Journal of Economics* 46 (3): 526–40. https://doi.org/10.2307/1883398.
Buchanan, James M., and Gordon Tullock. 1962. *The Calculus of Consent: Logical Foundations of Constitutional Democracy*. Vol. 100. University of Michigan press.
Charles, Loïc, Thomas Delcey, Yann Giraudc, and Iván Ledezma. 2025. "La Revue Économique (1950-2025), Une Perspective Quantitative:" *Revue Économique* Vol. 76 (3): 363–413. https://doi.org/10.3917/reco.763.0363.
Cherrier, Beatrice, and Jean-Baptiste Fleury. 2017. "Economists' Interest in Collective Decision After World War II: A History." *Public Choice* 172 (1-2): 23–44. https://doi.org/10.1007/s11127-017-0410-7.
Cherrier, Beatrice, and Andrej Svorenčík. 2018. "The Quantitative Turn in the History of Economics: Promises, Perils and Challenges." *Journal of Economic Methodology* 0 (0): 1–11. https://doi.org/10.1080/1350178X.2018.1529217.
Claveau, François, and Yves Gingras. 2016. "Macrodynamics of Economics: A Bibliometric History." *History of Political Economy* 48 (4): 551–92. https://doi.org/10.1215/00182702-3687259.
Cooley, Charles H. 1915. "The Progress of Pecuniary Valuation." *The Quarterly Journal of Economics* 30 (1): 1–21. https://doi.org/10.2307/1885801.
Culbert, Jack H., Anne Hobert, Najko Jahn, Nick Haupka, Marion Schmidt, Paul Donner, and Philipp Mayr. 2025. "Reference Coverage Analysis of OpenAlex Compared to Web of Science and Scopus." *Scientometrics* 130 (4): 2475–92. https://doi.org/10.1007/s11192-025-05293-3.
Cyert, Richard M., and Kenneth D. George. 1969. "Competition, Growth, and Efficiency." *The Economic Journal* 79 (313): 23–41. https://doi.org/10.2307/2229627.
Delcey, Thomas. 2025. "The Dissemination of Fama (1970): A Bibliometric Analysis."
Delcey, Thomas, and Francesco Sergi. 2023. "The Efficient Market Hypothesis and Rational Expectations Macroeconomics. How Did They Meet and Live (Happily) Ever After?" *The European Journal of the History of Economic Thought* 30 (1): 86–116. https://doi.org/10.1080/09672567.2022.2108869.
Desmarais-Tremblay, Maxime, Marianne Johnson, and Richard Sturn. 2023. "From Public Finance to Public Economics." *The European Journal of the History of Economic Thought* 30 (5): 934–64. https://doi.org/10.1080/09672567.2023.2248323.
Desrosières, Alain. 2011. *The Politics of Large Numbers: A History of Statistical Reasoning*. Translated by Camille Naish. Digital repr. 2011. Cambridge, Mass. London: Harvard University Press.
Downey, E. H. 1910. "The Futility of Marginal Utility." *Journal of Political Economy* 18 (4): 253–68. https://doi.org/10.1086/251690.
Downs, Anthony. 1957. "An Economic Theory of Political Action in a Democracy." *Journal of Political Economy* 65 (2): 135–50. https://doi.org/10.1086/257897.
Drazen, Allan. 1980. "Recent Developments in Macroeconomic Disequilibrium Theory." *Econometrica: Journal of the Econometric Society*, 283–306. https://doi.org/10.2307/1911097.
Duarte, Pedro G. 2025. "The Rise of the Rational Expectations Hypothesis." In.
Earl, Peter E. 2022. *Principles of Behavioral Economics: Bringing Together Old, New and Evolutionary Approaches*. Cambridge, United Kingdom New York, NY Port Melbourne, VIC New Delhi Singapore: Cambridge University Press. https://doi.org/10.1017/9781009091053.
Edwards, José. 2020. "Fifty Years of HOPE." *History of Political Economy* 52 (1): 1–46. https://doi.org/10.1215/00182702-8009485.
Edwards, José, Yann Giraud, and Christophe Schinckus. 2018. "A Quantitative Turn in the Historiography of Economics?" *Journal of Economic Methodology* 25 (4): 283–90. https://doi.org/10.1080/1350178X.2018.1529133.
Erickson, Paul, Judy L. Klein, Lorraine Daston, Rebecca Lemov, Thomas Sturm, and

Michael D. Gordin. 2013. *How Reason Almost Lost Its Mind – The Strange Career of Cold War Rationality*. Reprint. University of Chicago Press.
Friedman, Milton. 1953. "The Methodology of Positive Economics." In *Essays in Positive Economics*. Chicago University Press.
———. 1977. "Nobel Lecture: Inflation and Unemployment." *Journal of Political Economy* 85 (3): 451–72. https://doi.org/10.1086/260579.
Friedman, Milton, and L. J. Savage. 1948. "The Utility Analysis of Choices Involving Risk." *Journal of Political Economy* 56 (4): 279–304. https://www.jstor.org/stable/1826045.
Giocoli, Nicola. 2003. *Modeling Rational Agents: From Interwar Economics to Early Modern Game Theory*. Cheltenham, UK Northampton, MA: Edward Elgar Publishing.
Goutsmedt, Aurélien. 2021. "From the Stagflation to the Great Inflation: Explaining the US Economy of the 1970s." *Revue d'Economie Politique* 131 (3): 557–82. https://doi.org/https://www.cairn.info/revue-d-economie-politique-2021-3-page-557.htm?ref=doi.
Goutsmedt, Aurélien, François Claveau, and Catherine Herfeld. 2023. "Quantitative and Computational Approaches in the Social Studies of Economics." *Oeconomia - History, Methodology, Philosophy* 13 (2): 147–61. https://doi.org/https://journals.openedition.org/oeconomia/15799.
Goutsmedt, Aurélien, and Alexandre Truc. 2023. "An Independent European Macroeconomics? A History of European Macroeconomics Through the Lens of the European Economic Review." *European Economic Review* 158: 104559. https://doi.org/10.1016/j.euroecorev.2023.104559.
Grafton, Anthony. 1999. *The Footnote: A Curious History*. [Rev. ed.] 4. print., 1. Harvard Univ. Press paperback. Cambridge, Mass: Harvard University Press.
Grimmer, Justin, Margaret E. Roberts, and Brandon M. Stewart. 2022. *Text As Data: A New Framework for Machine Learning and the Social Sciences*. Princeton: Princeton University Press.
Grossman, Sanford J., and Joseph E. Stiglitz. 1980. "On the Impossibility of Informationally Efficient Markets." *American Economic Review* 70 (3): 393–408. https://www.jstor.org/stable/1805228.
Guldi, Jo. 2023. *The Dangerous Art of Text Mining: A Methodology for Digital History*. 1st ed. Cambridge University Press. https://doi.org/10.1017/9781009263016.
Hands, D. Wade. 2015. "Normative Rational Choice Theory: Past, Present, and Future." *Present, and Future (March 2015)*.
Herfeld, Catherine. 2018. "From Theories of Human Behavior to Rules of Rational Choice: Tracing a Normative Turn at the Cowles Commission, 1943–54." *History of Political Economy* 50 (1): 1–48.
Heukelom, Floris. 2014. *Behavioral Economics: A History*. Historical Perspectives on Modern Economics. New York, NY, USA: Cambridge University Press.
Hicks, J. R., and R. G. D. Allen. 1934. "A Reconsideration of the Theory of Value. Part I." *Economica* 1 (1): 52–76. https://doi.org/10.2307/2548574.
Hurwicz, Leonid. 1946. "Theory of the Firm and of Investment." *Econometrica* 14 (2): 109–36. https://doi.org/10.2307/1905363.
Hutchison, T. W. 1937. "Expectation and Rational Conduct." *Zeitschrift Für Nationalökonomie / Journal of Economics* 8 (5): 636–53. https://doi.org/10.1007/BF01311467.
Igersheim, Herrade. 2019. "The Death of Welfare Economics: History of a Controversy." *History of Political Economy* 51 (5): 827–65.
Jullien, Dorian, and Alexandre Truc. 2024. "Towards a History of Behavioural and Experimental Economics in France." *The European Journal of the History of Economic Thought* 31 (6): 998–1033. https://doi.org/10.1080/09672567.2024.2415000.
Kahneman, Daniel, and Amos Tversky. 1979. "Prospect Theory: An Analysis of Decision Under Risk." *Econometrica* 47 (2): 363–91.
Klaes, Matthias, and Esther-Mirjam Sent. 2005. "A Conceptual History of the Emergence of

Bounded Rationality." *History of Political Economy* 37 (1): 27–59. https://doi.org/10.1215/00182702-37-1-27.

Leibenstein, Harvey. 1966. "Allocative Efficiency Vs. "X-Efficiency"." *American Economic Review* 56 (3): 392–415. https://www.jstor.org/stable/1823775.

Lemercier, Claire, and Claire Zalc. 2019. *Quantitative Methods in the Humanities: An Introduction*. Translated by Arthur Goldhammer. Charlottesville: University of Virginia Press.

Lester, Richard A. 1947. "Marginalism, Minimum Wages, and Labor Markets." *The American Economic Review* 37 (1): 135–48. https://www.jstor.org/stable/1802863.

Lucas, Robert E, Jr. 1972. "Expectations and the Neutrality of Money." *Journal of Economic Theory* 4 (2): 103–24. https://doi.org/10.1016/0022-0531(72)90142-1.

Maas, Harro. 1999. "Mechanical Rationality: Jevons and the Making of Economic Man." *Studies in History and Philosophy of Science Part A* 30 (4): 587–619. https://doi.org/10.1016/S0039-3681(99)00030-8.

Machlup, Fritz. 1946. "Marginal Analysis and Empirical Research." *The American Economic Review* 36 (4): 519–54. https://www.jstor.org/stable/1801722.

Marschak, Jacob. 1950. "Rational Behavior, Uncertain Prospects, and Measurable Utility." *Econometrica* 18 (2): 111–41. https://doi.org/10.2307/1907264.

Martín-Martín, Alberto, Mike Thelwall, Enrique Orduna-Malea, and Emilio Delgado López-Cózar. 2021. "Google Scholar, Microsoft Academic, Scopus, Dimensions, Web of Science, and OpenCitations' COCI: A Multidisciplinary Comparison of Coverage via Citations." *Scientometrics* 126 (1): 871–906. https://doi.org/10.1007/s11192-020-03690-4.

Mitchell, Wesley C. 1910. "The Rationality of Economic Activity: I." *Journal of Political Economy* 18 (2): 97–113. https://doi.org/10.1086/251657.

———. 1914. "Human Behavior and Economics: A Survey of Recent Literature." *The Quarterly Journal of Economics* 29 (1): 1–47. https://doi.org/10.2307/1885296.

Mongin, Philippe. 2019. "The Allais Paradox: What It Became, What It Really Was, What It Now Suggests to Us." *Economics and Philosophy* 35 (3): 423–59. https://doi.org/10.1017/S0266267118000469.

Moretti, Franco. 2013. *Distant Reading*. London ; New York: Verso.

Moscati, Ivan. 2019. *Measuring Utility: From the Marginal Revolution to Behavioral Economics*. Oxford Studies in the History of Economics. New York: Oxford University Press.

Muth, John F. 1961. "Rational Expectations and the Theory of Price Movements." *Econometrica* 29 (3): 315. https://doi.org/10.2307/1909635.

Offer, Avner, and Gabriel Soderberg. 2016. *The Nobel Factor: The Prize in Economics, Social Democracy, and the Market Turn*. Princeton, NJ: Princeton University Press.

Periti, Francesco, and Nina Tahmasebi. 2024. "A Systematic Comparison of Contextualized Word Embeddings for Lexical Semantic Change." arXiv. https://arxiv.org/abs/2402.12011.

Reimers, Nils, and Iryna Gurevych. 2019. "Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks." arXiv. https://doi.org/10.48550/arXiv.1908.10084.

Rutherford, Malcolm. 2001. "Institutional Economics: Then and Now." *Journal of Economic Perspectives* 15 (3): 173–94. https://doi.org/10.1257/jep.15.3.173.

———. 2013. *The Institutionalist Movement in American Economics, 1918–1947: Science and Social Control*. Cambridge University Press.

Sargent, Thomas J, and Neil Wallace. 1973. "Rational Expectations and the Dynamics of Hyperinflation." *International Economic Review*, 328–50.

Savage, Leonard J. 1954. *The Foundations of Statistics*. New York: John Wiley.

Sent, Esther-Mirjam. 2004. "Behavioral Economics: How Psychology Made Its (Limited) Way Back Into Economics." *History of Political Economy* 36 (4): 735–60. https://doi.org/10.1215/00182702-36-4-735.

———. 2008. "Rationality, History of the Concept." In *The New Palgrave Dictionary of Economics*, edited by Palgrave Macmillan, 1–11. London: Palgrave Macmillan UK.

https://doi.org/10.1057/978-1-349-95121-5_2834-1.
Shackle, G. L. S. 1942. “A Theory of Investment-Decisions.” *Oxford Economic Papers*, no. 6: 77–94. https://doi.org/10.1093/oxepap/os-6.1.77.
Simon, Herbert A. 1955. “A Behavioral Model of Rational Choice.” *The Quarterly Journal of Economics* 69 (1): 99–118. https://doi.org/10.2307/1884852.
———. 1959. “Theories of Decision-Making in Economics and Behavioral Science.” *American Economic Review* 49 (3): 253–83. https://www.jstor.org/stable/1809901.
———. 1978. “Rationality as Process and as Product of Thought.” *American Economic Review* 68 (2): 1–16. https://www.jstor.org/stable/1816653.
———. 1979. “Rational Decision Making in Business Organizations.” *American Economic Review* 69 (4): 493–513. https://www.jstor.org/stable/1808698.
Sugden, Robert. 1991. “Rational Choice: A Survey of Contributions from Economics and Philosophy.” *The Economic Journal* 101 (407): 751–85. https://doi.org/10.2307/2233854.
Teplitskiy, Misha, Eamon Duede, Michael Menietti, and Karim R. Lakhani. 2022. “How Status of Research Papers Affects the Way They Are Read and Cited.” *Research Policy* 51 (4): 104484. https://doi.org/10.1016/j.respol.2022.104484.
Tobin, James. 1972. “Inflation and Unemployment.” *The American Economic Review* 62 (1/2): 1–18. https://www.jstor.org/stable/1821468.
Truc, Alexandre. 2022. “Forty Years of Behavioral Economics.” *The European Journal of the History of Economic Thought* 29 (3): 393–437. https://doi.org/10.1080/09672567.2021.1993295.
———. 2023. “Neuroeconomics: Hype or Hope? An Answer.” *OEconomia* 13 (-2): 239–77. https://doi.org/10.4000/oeconomia.15489.
———. 2025. “The Disciplinary Mobility of Core Behavioral Economists.” *Revue Économique* nº75 (6): 1091–1126. https://doi.org/10.3917/reco.756.1091.
Truc, Alexandre, Olivier Santerre, Yves Gingras, and François Claveau. 2023. “The Interdisciplinarity of Economics.” *Cambridge Journal of Economics* 47 (6): 1057–86. https://doi.org/10.1093/cje/bead021.
Tugwell, Rexford G. 1922. “Human Nature in Economic Theory.” *Journal of Political Economy* 30 (3): 317–45. https://doi.org/10.1086/253433.
Underwood, Ted, Laura K. Nelson, and Matthew Wilkens. 2025. “Can Language Models Represent the Past Without Anachronism?” arXiv. https://doi.org/10.48550/arXiv.2505.00030.
Veblen, Thorstein. 1909. “The Limitations of Marginal Utility.” *Journal of Political Economy* 17 (9): 620–36. https://doi.org/10.1086/251614.
Von Neumann, John, and Oskar Morgenstern. 1944. *Theory of Games and Economic Behaviour*. Princeton: Princeton University Press.
Winter, Sidney G. 1971. “Satisficing, Selection, and The Innovating Remnant.” *The Quarterly Journal of Economics* 85 (2): 237–61. https://doi.org/10.2307/1880703.
Yonay, Yuval Peretz. 2001. *The Struggle over the Soul of Economics: Institutionalist and Neoclassical Economists in America Between the Wars*. Princeton: Princeton University Press.
Zhang, Philipp, Jakub Rojcek, and Markus Leippold. 2025. “EconBERT: A Large Language Model for Economics.” {{SSRN Scholarly Paper}}. Rochester, NY: Social Science Research Network. https://doi.org/10.2139/ssrn.5263616.

Appendix

Overview of the data and methods

Here, a quick summary of all the databases used and the methods used. Could be nice to have a visual representation.

Building the full-text database

Making clear the difference between the bibliometric database and the full-text database, and the full-text database and the rationality corpus.

Elaborating the list of economics journals

Extraction and cleaning process for economics full texts

Overview of the full-text database content

Merging with the bibliometric database

Creating a rationality corpus

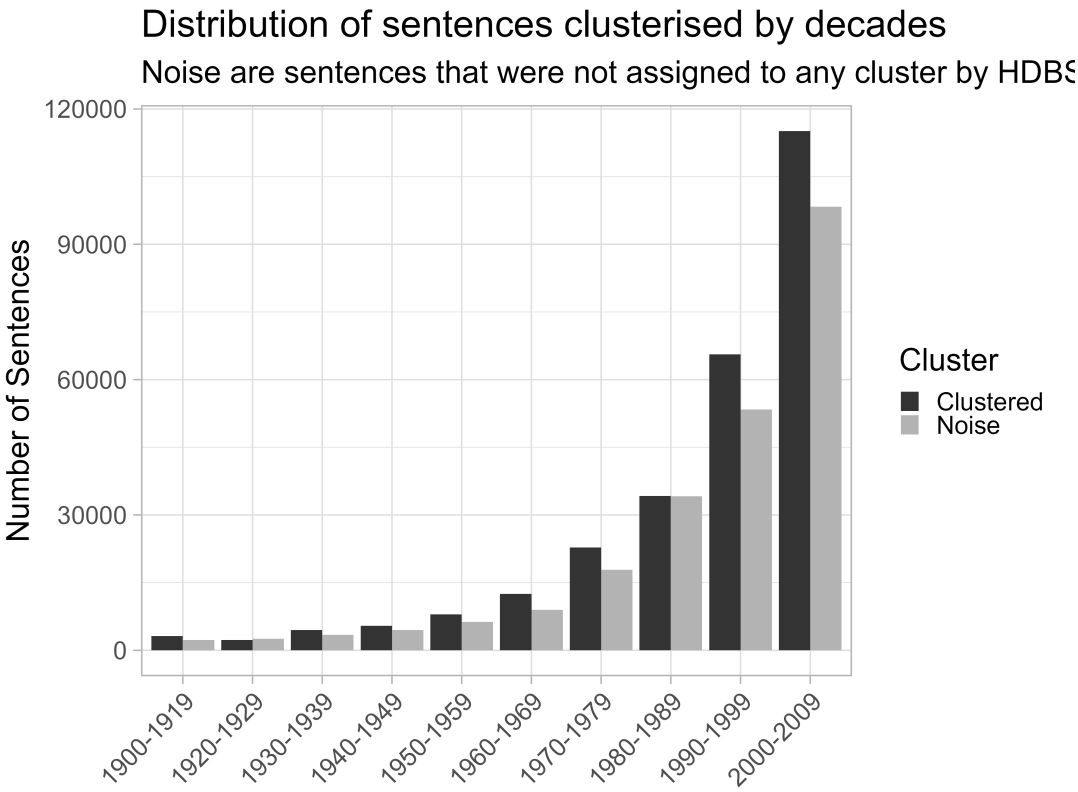
Using LLMs and sentence embeddings to represent sentences

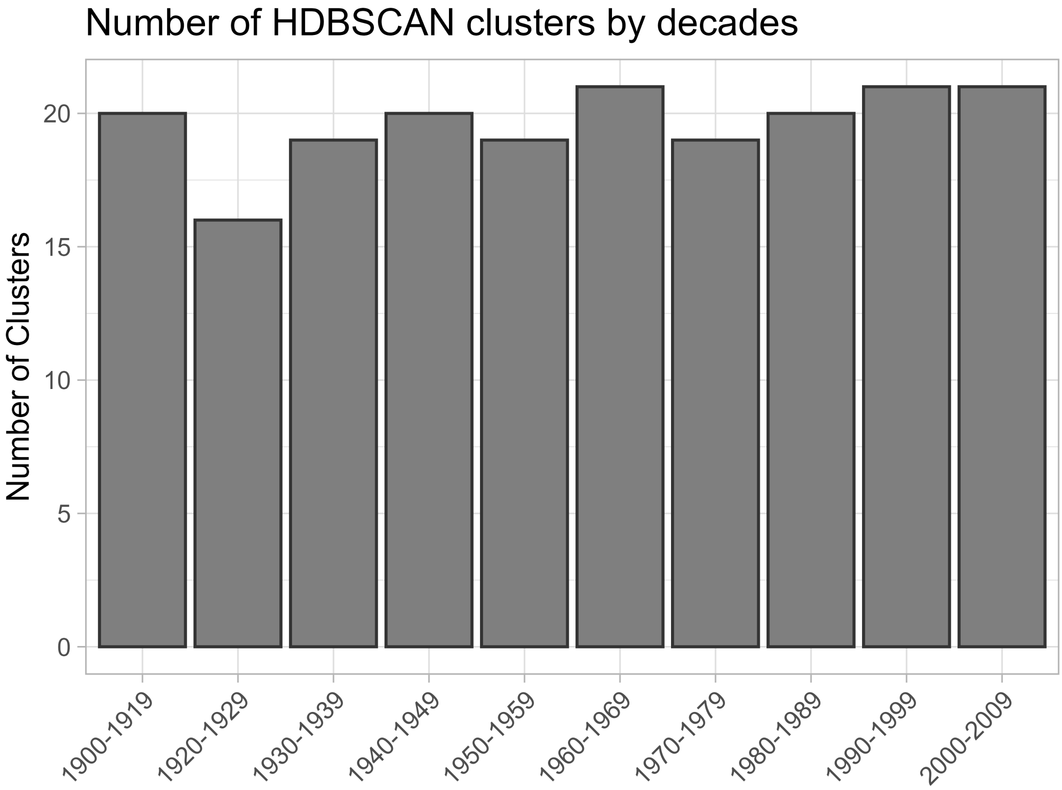
Building the representative vector of rationality

Selecting full-texts and sentences related to rationality

Text clustering methods

Sliding time windows and HDSBSCAN





Backbone networks and matching clusters over time

Bibliographic coupling